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2012/2013 Edition

Bestselling Guide

LAUTECH POST–UTME SCREENING
Past Questions & Solutions + 1 Year SMS Alerts

2011 - 2005

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PREFACE

Until the national consensus to adopt post-test by various tertiary institutions in the country, the Joint Admission and Matriculation Board-JAMB had remained the only body charged with the responsibility of testing, screening, preparing and presenting qualified secondary school leavers for admission into the over 76 universities (government and private inclusive). The introduction of the POST-UME is not unconnected with the malpractices associated in the conduct of the jamb exams nationwide in the past five years. Thus in 2010, the name degenerated from POST-UME to POST-UTME thereby incorporating and involving both Polytechnics and Monotechnics nationwide.

The advent of this "menace" as often seen by prospective Nigerian students was as old as 2005, today no school accepts candidates for admission without duly testing the person's ability through the post Jamb test. This therefore possess yet another hurdle and barricade to young school leavers; bridging them from getting admitted each year.

In the course of, and in preparation for this examination/test, the prospective student faces a lot of worries like; how many questions am I expected to write? What schemes and subjects do I need to meet up with? When will the application start? How am I going to be judged or graded? What are my chances of getting admitted? And of course where can I get a prototype, copy and replica of 'my choice of school' past questions and possible solutions to the questions? All these and more had remained rhetorical until the emergence of MYSCHOOLCOMM ADMISSION SUPPORT.

In our quest to remain, maintain and sustain our current status as the best and ever first corporate organization that manage the entirety of Nigerian Undergraduates, we had in the past one year embarked on massive compilation, revision, analysis and research on previous post UTME examinations conducted, the feasibility studies and chances of a candidate passing or and of course failing the test. From our research findings, we have found without further verifiable hypothesis that the POST-UTME past questions remain a secret to passing through these tests in success.

We therefore present and recommend these past questions for anyone/everyone preparing for this examination.
Our idea in this sense does not focus on duplicating the efforts made by several other Nigerians, Lecturers, teachers and tutors in this direction. We only aimed at correcting some of their shortfalls, and most importantly; making these study, revision and preparatory materials readily available to anyone in any part of the country.

The questions contained herein have been drawn from the candidate's school of interest/choice in the relevant courses to facilitate easy and more result orientated performance. In this respect, actual and real examination questions were extracted from your school of choice and are therefore being presented to you, read well to succeed.

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LADOKE UNIVERSITY OF TECHNOLOGY

2006 POST UME SCREENING TEST

Read this passage

Over the years, time series forecasting has been existence to make timely decision in the face of uncertainty about the future. The making of forecast is an essential what aspect of our life. Basically, all the of forecasting work or the premise that if we can predict that the future will be like we can modify our behaviors or the behavior of the system for better position.

ANSWER THE FOLLOWING QUESTIONS.

1. From the passage. We gather that forecasting is
   (a) new science [b) an old science [c) an unknown science [d) a future science
2. Forecasting become necessary because of
   (a) the need to make timely decisions (b) its importance to our life
   (c) unpredictability of man's nature (d) fear of unknown
3. The expression, "over the years" in the passage above is an example of
   (a) Adjectival phrase (b) adjectival clause (c) noun phrase (d) adverbial phrase
4. According to the passage, if forecasting helps to deal with an uncertainty future, it can the best be describes as
   (a) an intervention science (b) a preventive science
   (c) a corrective science (d) a dialogist science
5. The under listed expression in the passage is grammatically known as
   (a) noun clause (b) prepositional phrase
   (c) conditional clause (d) adjectival clause

Read the following passage and answer the question that follow

A green desire perfumed memories, a leafy longing by my under feet to this forest to a thousand wonders.
A green desires for this petalled umbrella of simple star-sand compound suns. Suddenly, so the sky is free high

6. The scenario the passage above suggest
   (a) a busy street (b) an unfolding tragedy
   (c) an agrarian environment (d) a school set-up
7. The last line of the text hits at
   (a) a raging storm (b) an impending rain (c) a very tall tree(d) a thunderous cloud
8. The expression, "petalled umbrella in the passage means
   (a) High colored canopies (b) dense forest foliage
   (c) a kingly umbrella (d) a nice-looking cap.
9. According to the passage, the writer finds himself at the scene be described out of
   (a) curiosity (b) compulsion (c) feeling (d) memory
10. An appropriate title for this passage
    (a) celebration (b) memories (c) forest tunes (d)Nature's echoes

Choose the right option that best completes each of the following:

11. He didn't know anything about business, so starting his own was
(a) a leap into the cloud (b) a leap in the dark (c) a leap into the ocean.

12. I like the way he criticizes everybody. It really rattles
(a) my back (b) my bones (c)my heart (d) my cage

13. When her business crashes. She had to pick up and start …………….again
(a) the fragment (b)the losses (c) the pieces (d) the stones.

14. She felt really bad when she realized that she had lost her watch. It wasn't expensive but it had sentimental
(a) value (b) price (c) cost(c)expense

15. I used to go to church under false ……………I never wanted to go but my mother made me.
(a)agreement(b) feeling (c) pretences(d) façade

Choose the option opposite meaning to word or phrase in italics in question 16-20

16. The university senate building is rather gigantic
(a) roomy (b) small (c) huge (d) thin

17. Mama Tolu has really become quite chubby
(a) Insomnia (b) stubborn (c) thin (d) round

18. She deliberately killed the goat
(a) carelessly (b) mercilessly(c)brutally(d)unintentional

19. The former president of the association received a derogatory remark from the member (a) receive standing ovation
(b) received public opprobrium
(c) received encomium (d) received bitter Ruth

20. Adebomi was dismissed from dereliction of duty
(a) insubordinate (b) irresponsibility(c) laziness (d) negligence

In each of question 21-24. Choose the same consonant sound as the one represented by the letter underlined.

21. Throb (a) Those (b) These (c) Thousand(d)This
22. Judges (a) crush (b) crunchy (c) garage(d) gasp
23. Quay (a) key (b) Queen (c) Try (d) pry
24. Unique (a) church (b) Arch (c) Architect (d)charade

In each of question 25 below choose the option that is nearest in meaning to each of the underline word or phrase in each sentence

25. The idea expressed by the chairman of the party is Make believe
(a)The idea is good(b)the idea is impressive(c) the idea is well-conceived

ANSWER KEY

1. B	2.D	3 C	4 C	5 —
6 C	7. C	8 B	9D	10D
11C	12B	13.C	14 A	15 C
16 B	17 C	18 D	19C	20 A
21C	22 C	23 A	24C	25 C

LADOKE AKINTOLA UNIVERSITY OF TECHNOLOGY

2007 POST UME TEST

Each of questions 1-5 below, choose the following that is nearest in meaning to each of the underlined word or phrase in each sentence.

1. The idea expressed by the chairman of the party is make— believe
   (a) the idea of good (b) the idea is impressive
   (c) the idea is well conceived (d) the idea is a form f pretence.
2. It rained cat and dog
   (a) it rained mercilessly (b) it rained drastically (c) it rained heavily (d) it rained pieces.
3. Bolanle was taken on as graduate assistant in the university
   (a) Bolanle was laid off (b) Bolanle was denoted
   (c) Bolanle was employed (d) Bolanle was promoted.
4. Toe's emphalate disposition to life has really recorded her the chance to become the Minister of Internal Affairs
   (a) expressive mannerism (b) coolative outlook of life
   (c) internal And friendly way of life (d) spent able.
5. Biola managed to hold her head high and ignore what people were saying about her at the conference
   (a) Biola was rude to people (b) Biola was able to display her affluence
   (c) Biola was discouraged (d) Biola was proud and obstinate.
6. It was calculated mutiny to dethrone the newly installed sultan of Iyanfoworogi
   (a) act of revival (b) act of loyalty (c) act of facility (d) act of rebellion.
7. The objectives of the work are well — delineated
   (a) well - done (b) well described and explained (c) well - focused (d) well - drawn.
8. The university review committee comment that Mr. Ojongbolo's vitae leaves must to be desired
   (a)highly commendable (b) highly recommended
   (c)highly approved (d) highly unacceptable.

ANSWER KEY

1. C 2.C 3.C 4. C5D6,D 7. B 8.D

LADOKE AKINTOLA UNIVERSITY OF TECHNOLOGY 2006 POST UME TEST

I. Which of the following does not contribute to the biomass in an ecosystem.
   a. Producers b. Food chain c. Consumers d. Saprophytes
2. Terrestrial animals which are capable of mintai fling constant body temperatures within fairly close limits are.
   a. Thermocines b. Homotherms c. Poikilotherms d Sternotherms
3. The amount of moisture in the air (relative humidity) can be measured by
   a. hydrometer b. anemometer c. rain guage d. hygrometer
4. Which of the following groups of plants would be the first colonies in an ecological succession on changing rocks to soil?
   a . Mosses b. Ferns c. Lichens d. Grasses
5. Plants adapted to life is salty water are called
   a. hydrophytes b. epiphytes c. halophytes d. xerophytes

6. The state in the life of bilharzias which infects man is the
   a. cercaria b. bladder worm c. miracidium d. Egg
7. The malphigian tubule plays a major role in
   a. inspiration b. excretion c. secretion d. digestion
8. The filtered blood from the kidney is carried back to the circulatory system, through the
   a. hepatiê portal vein b. pulmonary vein c. renal vein d. renal artery
9. Flatworms are classified as
   a. Concentrate b. Annelid c. Playhelmintie d. Arachinida
10. A severe deficiency of thyroxin result in
    a. sexual underdevelopment b. cretinism c. gigantism d. diabetes mellitus
11. Haemolysis is an example of
    a. Hydrolysis b. Osmosis c. Plasmolysis d. Inhibition
12. Which of the following structures are visible in the cell of a mitosis
    a. Centrioles, chromatides and nucleolus
    b. Homologous chromosome, nuclear membrane
    c. Cell wall, centrioles and chromatids
    d. Chromosomes, nuclear membrane and centromere
13. At which of the following stage of cell division can the cell resting?
    a. Anaphase b. Telophase c. Prophase d. Interphase
14. Mitochondria and chloroplasts are characteristic of cells that
    a. Reproducing and photosynthesizing
    b. Excreting and deaminating
    c. Respiring and photosynthesizing
    d. Replicating and photosynthesizing
15. Which of the following organelles help to remove excess water from cell?
    a. Mitochondria b. Ribosome c. Contractile vacuole d. Golgi body
16. Which of the following tissues is made up of dead cells
    A. Meristem B. Cambium C. Xylem D. Phloem
17. Marriana handle in a monocotyledonous stem are found
    a. Watered in the epidermis
    b. The ring near the epidermis
    c. Between the condoderms and pericycle
    d. Scattered between the epidermis and the pith
18. Plants that can survive in an environment where water is scarce are referred to as
    a. Epiphytes b. Mesophytes c. Hydrophytes d. xerophytes
19. Green plant are primary producers because they are
    a. Heterotrophic b. Saprophytic c. Chlorophytic d. Autotrophic
20. The hypha of Rhizopus is said to be conencytic because it
    a. does not contain chlorophyll b. has no cross walls
    c. is vacuolated d. stores oil globules
21. The skeleton of an anthropod is principally composed of
    a. pectin b. chitin c. lignin d. tannin
22. The aerobic stage of tissue respiration occurs through
    a. Energy cycle b. Kreb's cycle c. Water cycle d. Carbon cycle
23. Extracellular digestion take place in

a. Green plants c. Algae d. Bryophytes

24. Which of the following reagents is used to test for starch
a. Million's test b. Fehling's solution c. Sudan III solution d. Iodine solution

25. The condition known as cretinism is caused by the deficiencies
a. Adrenalin b. Vitamin A c. Insulin d. Thyroxin

26. The key event in the transition of the amphibians from water to land is
a. possession of we skin b. possession of webbed limbs
c. replacement of gills with lungs d. development of lung hint limber

27. A group of organisms that are capable of free interbreeding is called
a. species b. genus c. family d. order

28. Which of the following organisms has only two body layers and a cavity?
a. Hydra b. Taenia Spp c. Ascaris d. Earthworm

29. Which of the following scientists was involved with organic evolution?
a. Mendel b. Lamark c. Morgan d. Luther

30. Poliomyelities is a microbial disease cause by
a. Amoeba b. Protozoa c. Virus d. Protists

Answer Key

1. D	7.A	13.D	19.D	25.D
2.B	8. C	14.C	20 -	26. C
3.D	9.C	15.C	21.B	27.A
4.C	10. B	16.C	22. B	28. A
5.C	11.B	17-	23.A	29.B
6.6	12.B	18.D	24.D	30C

Explanation to the question

1. Biomass takes into accounts both the size of the individual organisms and their numbers. The pyramid of biomass thus gives the accurate picture of the relationship between the organisms at the various tropic level in a food chain is just a mode of nutrition and it's not relevant in this context (D).
2. Homiotherms (Greek: homois: similar) animals keep their body temperature more or less constant, regardless of their surrounding (B)
3. Relative humidity is the measurement of the amount of moisture in the air and is determined by using a wet and dry bulb hygrometer (D).
4. Lichen is as a result of symbiosis relationship association between fungi and algae plant. It is the first stage of ecological succession. Lichen is the simplest of the options. (C).
5. Hydrophytes or aquatics live in fresh water and show a number of adaptation to their environment. Xerophytes are plants which can survive in place where water supply is limited. Epiphytes are involved in commensalism. Halophytes means plant in halides environment that is contain salts of chloride (C).
6. Miracidium larvae emerges and bores into foot of small Redia forms further larvae by internal propagation. Cercavia escapes from snail through pulmonary aperture. Encysted cercaria adheres to blade of grass in wet pasture land and get into man. (A)
7. Malphigian tubule is the excretory organ in insect (A).
8. The blood is transported to the Kidney through the artery and leaves the Kidney through Renal vein (C).

9. Flatworms are platyhelmintlies (C).
10. Thyroxine is complex compounds which include iodine. An animal not producing enough becomes slow, may not grow to its proper size (cretinism) (B).
11. If a cell is placed in a weaker (i.e. hypotonic) solution, they swell and may even burst. The phenomenon is known as haemolysis. Just like osmosis which in the movement of solvent molecule of high concentration to lower concentration through semi — permeable membrane.(B).
12. At the interphase, the chromosome are not visible as distinct bodies, either under light microscope or electron microscope at this stage, they form a long chromatin strands or threads. Not until prophase where the chromatin threads condense to form visible chromosome. The nucleolus disappear at the late prophase stage. Note: check textbooks. (B).
13. The interphase is described as Resting stage but it is a complete misnormer because the genetic material replicate and cell builds up at this stage (D).
14. Mitochoiidria is the power house, food material is oxidized to give energy while chioroplast is involved in photosynthesis(C).
15. Contractile vacuole helps in osmo regulation that is, regulation of amount of water in the cell. Ribosome is for protein synthesis.(C).
16. Meristem is the tissue capable of cell division. The xylem contains several large thick walled tubes called vessels in which the soil solution is carried from the root to the leaves, they do not contain cytoplasm i.e. they are dead. (C)
17. -
18. Xerophytes are plants that adapts to desert habitat (D).
19. Green plant are anthrotrophic, they synthesize their food themselves (D).
20. -
21. Chitin is a proteinous material that made up exoskeleton in insects(Arthropod) (B).
22. In terms of ATP — synthesis Kreb's cycle is the most important source of energy in metabolism and it achieves this by donating hydrogen atom(s) to the carrier system. This takes place during respiration (B).
23. Moulds are saprophytes, they feed on the dead bodies of plant or organic materials like bread. They therefore pass out enzymes from their cells on to the substrate (the material in which they feed) to make it soluble, this kind of digestion occurring outside the organisms called Extracellular digestion (A).
24. Starch turns iodine solution blue — black (D).
25. (D) See Qun 10
26. Amphibians use gill for gaseous exchange in aquatic environment while the gill is changed to lungs in Terrestial habitat (C).
27. A.
28. Diploblastic animals have two cell layer in their body wall, an outer layer (ectoderm) and an inner endoderm e.g Coelentrates (Hydra) (A).
29. Larmack's theory states that when an organism develops a need for a particular structure, this induce its appearance (B).
30. C.

LADOKEAKINTOLAUNIVERSITY OF TECHNOLOGY2010 POST UME TEST

1. The number of electrons in the K. L. M and N — shells of calcium are respectively

a) 2,8,8,2 b)2,2,8,8c) 2,8, 2, 8, d) 8,8,2,2

2. The basic unit of synthetic rubber is

a) Isoprene b) pentane c) butadiene d) butene

3. What are the oxidation numbers of manganese in the anions $Mn0_2$ and Mn04?

a) +7&+6 b)+2&+2 c) +4&+4

4. The empirical formula of a hydrocarbon that contains 93.3% carbon is C 12, H = 13

a) CH b) CH_2 c) C_2H d) C_2H_2

5. The type of hybridization in all the carbon in saturated hydrocarbon is

a) SP^3 b) SP^2 c) SP d) PS^2

6. Electrophiles are

a) Electron deficient species b) Electron rich species

c) Free radical species d) Negatively charged species

Explanation to Questions

1. K-shells contain maximum of 2-electron, L shells can only contain maximum of 8 electrons, M-shells contain maximum of 18 electrons, N- contains 32 electrons. Calcium with atomic number 20, shows that electrons are added to a shell until a stable duplet (for the K-shell) and an octet for L-shells. The increase in the number of electron result in the electrons occupying the shell with next highest energy level so calcium has 2, 8, 8, 2 (A).
2. Synthetic rubber is a polymer and the simple unit (monomer) is 2-methylbuta- 1, 3-diene known as isoprene unit (A).
3. MnO^{2-} = -2

X+(-2x2) =-2

X—4 = -2

X = +2

Mn04 = -1

X+(-2x4)=-1 -

X = -1+8

X=+7 (B)

4.

	C	H
% composition Mole ratio	92.3%	77
$^{92.3}/_{12}$ $^{7.7}/_{1}$	7.67	7.7
Mole ratio of element	7.67	7.7
Smallest mole ratio	7.67	7.67
=	1	1

Empirical formula CH (A)

5. Saturated hydrocarbons have single bond around its carbon, so they exhibit SP3 hybridization (A)
6. Electrophiles are electron deficient species like $NH4^+$,H^+ (A)

LADOKE AKINTOLAUNIVERSITY OF TECHNOLOGY 2007 POST UME TEST

1. The number of individual in a habitat is release to the unit space available to each organism is referred to as the

a. Birthrate b. Frequency c. Mortality d. Density

2. The group of bacteria that are involved in the conversion of ammonia to nitrate is

a. Antibacterial b. Neurosepoys c. Thizobium d. Clostridium

3. The sequence of ear ossicles from the sense ovalis

a. Malleus, incus and stepes b. Malleu, stepes and incus
c. Stepes, incus and malleus d. Stepes, malleus and incuse

4. The major function of the cell membrane is that
a. Delimits the cytoplasm b. Synthesis protein
c. Breakdown spindles d. Is the sites of photosynthesis

5. The network of double membrane that conveys material through the cytoplasm
a. Plasma membrane b. Vaccine membrane
c. Nuclear membrane d. Golgi body membrane

6. In the plants exhibiting alternation of generation the diploid multicellular stage known as
a. Sporoophytes b. Gametophyte c. Hetaphyte d. Cimatiophyte

7. Secondary thickening is initiated in the dicotyledous stem by the
a) Xylem parenchyma b) Secondary phloem c) Endodermis d) Cambium

8. The increase in the width of blood vessels the mammalian skin at high temperature known as
a. Constriction b. Vasodilation c. sweating d. excretion

9. The genotypic ratio of 1:2:1 in the offspring of a hybrid cross illustrates the law of
a. use and disuse b. dominance c. segregation d. variating

Answer Key

1. D	4.A	7.D
9. D.		
2. -	5.D	8.B
3.A	6.A	9.D

Explanation to the Answers

1. Population density is the average number of individuals of a species per unit area of the habitat (D).
2. No correct Option. Nitrosomonas first convert ammonia to Nitrites (N0,)while Nitrobacter completes the conversion to Nitrates (N03)
3. The middle ear consists of the eardrum (tympanium) and three small bones called ossicles which are malleus (Hammer), incus (anvil)and stapes (stirrup)in that order, that function in transmitting vibration. Stapes then link up with fenestral ovalis (A).

4. Cell membrane envelop the cytoplasm (A).
5. Golgi body is a stack of flat, membrane — bounded sacs. These protein filled sacs migrate to the surface of the cell and discharge their contents to the outside (D).
6. In moss plant, which exhibits alternation of generation. The sexual organ is called haploid gametophyte and asexual organ called diploid sporotophte. (A).
7. In the dicotyledon roots and stems, cambium exist between the xylem and the phloem. They are capable of living and multiplying thereby xylem and phloem. This then results in the growth in width or growth of the stem called secondary thickening (D).
8. The mechanism is to increase the rate of heat loss to the environment during dry season (B).

LADOKEAKINTOLAUNIVERSITY OF TECHNOLOGY 2008 POST UME TEST

1. During the process of starch formation in the leaves, the oxygen given of is desired from
   a. Sunlight b. Chlorophyll c. carbon dioxide d. Water.
2. Which of the following organs regulates the amount of amino acids and glucose in the body?
   a. Liver b. Kidney c. Pancreas d. Spleen
3. When two organisms heterozygous for a tract are made to cross, the phenotypic ratio of the offspring produced will be
   a. 2:1 b. 1:2:1 c. 3:1 d. 1:2
4. Glycolysis is best described as
   a. Splitting of glucose in the presence of oxygen
   b. Fermentation of starch
   c. Splitting of glucose in the absence of oxygen
   d. Formation of carbon dioxide and water from glucose
5. Which of the following is not true?
   a. Inhaled air contains more carbon dioxide that exhaled air
   b. Exhaled air contains more carbon dioxide than inhaled air
   c. Exhaled air is warm and contains water vapour
   d. Movement of diaphragm aids respiration.
6. Partially digested food ready to leave the stomach is referred to as
   a. Curd b. Chyme c. Glycogen d. Roughage
7. Mosses, Liverworts and ferns can be grouped together because they
   a. are all aquatic plants b. are grow in deserts
   c. are seedless plants d. all produce colorless flowers
8. In spirogyra, the pyrenoid
   a. Excretes waste products b. Is mainly used for respiration
   c. Is used to store starch d. Is suspended by cytoplasmic strands
9. Viruses are regarded as non — living because they
   a. Can neither reproduce asexually nor sexually
   b. Can not survive in their respective environments
   c. Do not posses characteristics transmutable from one generation to the next
   d. Can neither respire nor excretes.

ANSWER KEY

1.D	4. A	7. C
2.A	5.A	8. C
3.C	6.C	9. B

EXPLANATION TO ANSWER

1. $2H_20 \longrightarrow 4H^+ + 4OH$
   $4OH^- + \text{Chlorophyll} \longrightarrow 2H_2O + O_2 + \text{Energy}$ (D)

2. Kidney is an excreting organ and acts as osmo-regulators. Liver is a large organ lying under the diaphragm. One of the important functions of the liver is to regulate the amounts of substances passed into the general circulation. Excess sugars are temporarily stored in the liver as animal starch or glycogen (A).

3.

	T	t
T	Tt	Tt
t	Tt	tt

3 tallness: 1 shortness (C)

4. In the first stage of respiration, sugar is broken down, step by step, to pyruvic acid, a compound containing three carbon atoms. This sequence of reaction is known as glycolysis (A).
5 . Human being inhaled air from the environment, the larger part of the air is oxygen and after different process (respiration), human being exhale gas which is majorly carbon dioxide (A).
6. The wall of the stomach are muscular and regular peristaltic movement chum up the food, mixing it thoroughly with the gastric juice. By the time it is ready to leave the stomach the food looks like a watery paste called CHYME (C).
7. They are seedless plants (C).
8. Pyrenoid is for starch storage (C)
9. Although they have no means of propelling themselves they can reproduce and have characteristics which are transmitted form one generation to the next. They can only replicate inside the living host (B).

LADOKEAKINTOLAUNIVERSITY OF TECHNOLOGY2009 POST UME TEST

1. In bird, the following feathers possess after shaft
a. Quill and filoplumes b. Down and filoplumes
c. Covert and down d. Quill and covert
2. The nutritive layer of the eye in mammals is
a. Refracting media b. Conjunctiva c. Cornea d. Sciara
3. Ultra filtration in the Kidney takes place in the
a. Bowman's Capsule b. Pelvis c. Loop of Henle d. Proximal Convulated Tubule
4. Which of the following bones is not a component of the fore limb?
a. Olearanon b. Ulna c. Tibia d. Humerus
5. The condition in which the anthers mature before the stigma is called.
a. Protandry b. Epigyny c. Hypogyny d. Protogyny
6. In most true ferns, sporangia are grouped into
a. Indusium b. Fronds c. Son d. Prothalls
7. The ratio of carriers to sucklers in the F2 generation derived from a parental cross at two carriers of haemoglobin S gene is
a. 3:1 b. 1:3 c. 2:1 d. 1:2
8. In which part of a legurninous plant can bacteria like Azotobacteria be found ?
a. Spongy mesophyll b. Root nodes c. Stern internodes d. Stem nodes

9. In a dicotyledonous stem, companion cells are found close to the
   a. Endodermal cells b. Silver tubes c. Xylem vessels d. Pericyclic fibres
10. The position occupied by an organism In a food chain is referred to as
   a. Trophic level b. Niche level c. Energy level d. Feed level

ANSWER KEY

1. C	5.A	8. B
2. -	6. B	9. B
3.A	7. C	10.A
4. C		

EXPLANATION TO THE ANSWERS

1. The possession of feather is the most obvious difference between birds and other vertebrates. At the base of the vane, the barbules do not interlock and the barbs form a small tuft called the after shaft. Convert and down possession shaft (C).
2. The middle layer (choroid coat). Choroid Is pigmented and contain many capillaries and it's the nutritive layer of the eye (no correct options).
3. It was found that the fluid in the capsular space has almost exactly the same composition as blood plasma means the plasma proteins. It seems that the capsular fluid is formed by a process of ultra — filtration from 'the glomerular capillarus (i.e Ultra — filtration takes place in the Bowman's capsule) (A).
4. Ulna and Radius articulates at the elbow with the humerous at the fore limbs. Tibia belongs to the hind — limbs (C).
5. In protandry, the stamens (Anthers) ripen first and shed their pollen before the stigma
   Is matured (A).
1. At certain seasons, large number of spore — bearing bodies (sporandia) appear on the under surface of mature fronds. At first, theses are green, but they become brown as they mature. The sporangia are gathered into groups over the veins. Each group is called a sorus and each has a cover called an Indusium (B).

2.

	A	S
A	AA	As
S	As	SS

The ratio of carrier (AS) to Sucker is 2:1 (C).

8. Azobacter is found at the root nodules of Leguminous plant (B).
9. Companion cells are narrow cells density filled with cytoplasm. There is one beside each sieve tube (B).
10. A food chain shows thee transfer of energy and nutrients from organism to organisms in a feeding pathway. A food chain involves at least two links and more organisms at the lower tropic level to the higher tropic level (A).

LADOKEAKINTOLAUNIVERSITY OF TECHNOLOGY

2009 POST UME TEST

Instruction: Read the passage below and answer the question that follows

Epilepsy is a condition in which the patient is subject to recurrent attacks of loss of consciousness. Known as 'fits' on-losing consciousness, the patient falls and may hurt himself, and though in some cases the fit may end at this point, most attack go on to a stage in which the muscle on the body became rigid and the breathing is interrupted. This in turn is usually followed by the convulsive stage, in which there are jerking movements of the head, limbs and hand. The tongue may be bitten and the patient's writhing, irregular breathing, starring eyes and blue lips may be very alarming to the spectator. Then the patient slowly recovers, though when consciousness is fully restored the patient is still in a weakened condition and suffers unpleasant after-effects. Onlookers often insist on calling an ambulance but this is unnecessary. The patient should be made as comfortable as possible and allowed to rest.

Questions:

1. As used in the passage, the word convulsive' is a\an
   a. nominal phrase b. adjective c. gerund d. Preposition
2. The word, 'after-effects' in the fifth sentence of the passage is an example of
   a. compound noun b. adjective c. gerundial marker d. prepositional phrase
3. The word, 'spectator 'can also be used in the context of
   a. sport b. academics hospital interaction d. dancing

Section B

Instruction: Choose the correct option to fill the gap from the list under each of the sentences

4. Sales ________during the festival period
   a. peeked b. peacked c. picked d.peekid
5. It is only a fool that suffers in the _______of the plenty
   a. mist b. midst c. midct
6. We promised _____to make
   a. amend b. amend c. amends d. a mends
7. The man divided the job between you and
   a. 1 b. myself c. me d. myself

Instruction From the words or groups of words letter A-D, choose one which is mostly nearly opposite in meaning to the under listed expression as it is used in each of the following sentences:

8. Alhaji Taju Ologbenla is a prosperous businessman
   a. unsuccessful b. unskillful unscrupulous d. unskilled
9. All the must pass the compulsory subject
   a. Unimportant b. optional c. unreliable d. inferior
10. The security man acted courageously when thieves attacked the bank,

a. indiscreetly b. fearlessly c. shyly d. timidly” instruction:

ANSWER KEY

1.B 2.D 3.A 4.C 5.C 6.B
7.A 8.A 9.B I0.D

LADOKE AKINTOLA UNVIERSITY
OF TECHNOLOGY
2010 POST UME TEST

1. The man who killed the dog is here. The underlined in the above sentences is an example of
   (a) A phrase (B) A clause (c) an adjective (D) adverbial phrase

From the words letter A-D, choose the options that best complete the sentences.

2 The man acceded _________the Child’s demands
   (a) for (b) in (c) of (d) to
3. The weather was accurately
   (a) guessed (b) forecasted (c) fore cast (d) predicted

From the list of words or group of words lettered A to D below, choose the word or group of words that the sentences.

4. The polices man stopped all the _________to check and arrest all tax defaulters.
   (a) passer — by (b)passers — by (c) passer — byes (d) passer — byes

Pick the words that are opposite in meaning to the underlined expressions as used in these sentences.

5. The team played a lack luster game
   (a) dull (b) brilliant (c) enthusiastic (d) timid
6. Grandma is known to be ponderous
   (a) snobbish (b) instinctive (c) impulsive (d) gentle

(Use the short passage below to answer Question 7.

The word “paragraph” means writing ordinary or drawing with light. Therefore, a picture is one drawn with rays of light. Essentially, the camera box is a box with an aperture at one end of it .The aperture allows high into the camera box as a window does for a room

7. The percentage representation of nouns in the above passage
   (a) 19% (b)32% (c)44% (d) 14%
8. An adjective is always positioned
   (a) after prepositions (b) before the noun
   (c) after the noun (d) at the beginning of a sentences

9. The logical arrangement of sentences in a paragraph is known as
(a) clarity (b)completeness (c) coherence (d) unity
10. The word class comprises of the following except
(a) prepositions (b) phrases (c) pronouns (d) demonstrative

ANSWER KEY

1. A 2. D 3 C 4 B 5 C 6C
7. C 8 B 9 C 10 D

LADOKEAKINTOLAUNIVERSITY OF

TECHNOLOGY2010 POST UME TEST

1. All these are components of blood except
a. Plasma b. Erythrocytes c. Leucocytes d. Fibrin
2. When the F1 of a cross between two parents with pair of contrasting characters is selfed, the F2 population segregates approximately in a ration of……………………………………
a. 2:1 b. 1:1 c. 1:2 d. 3:1
3. The specific role played by an organism in its environment is known as……………………..
a. its function b. its responsibility c. Ecologicalniche d. its expected role
4. Which of these is a common property of cell ecosystems?
a. flow of energy b. decomposition of organic matter
c. Energy flow and nutrient cycling d. presence of plants and animals
5. Darwin theory of evolution explains the origin of species based on the existence of --
a. Variations b. Dominant characters c. Genes d. Acquired characters
6. All these are true about chloroplast and mitochondria except
a. they both have inner membrane reticulations
b. they both have their own DNA's
c. They are double membrane organelles
d. they are found in plant and animal cells
7. The following structures are used for breathing in toad except……………
a. Buccal cavity b. Lung c. Skin d. alimentary system

ANSWER KEY

1.D 2.D 3.C 4.D 5.B 6.D 7.D

EXPLANATIONS TO THE QUESTIONS

1. Erythrocytes, Leucocytes and plasma are all components of blood but there is a fascinating process in which more than 12 — different chemical factors present in the blood bring about the conversion of the soluble plasma protein fibrinogen into a mesh work of fine fibres called Fibrin (D).

2. (3:1) (D)

	T	t
T	Tt	Tt
t	Tt	tt

3. Niche is a place which is suited to the way of life of an organism within a community herbivore and a carnivore may share habitat but their different feeding methods means that they occupy different niches (C)
4. Ecosystem is an ecological unit made up of a community of all plant and animals living in a habitat and the non — living part of the environment (D).
5. Charles Darwin is most closely associated with the theory of evolution Darwin has an unparalleled opportunity to explore the flora and fauna in many different part of the world. During his long voyage, Darwin was forced further towards the conclusion that animals and plants have evolved by a process of slow and gradual change over successive generation. (B).
6. Chloroplast is absent in animal cell (D).
7. Toad exchange gas through lung, skin and buccal cavity. Alimentary system is involved in digestive system (D).

GENERAL KNOWLEDGE

1. Democracy day is celebrated in Nigeria on
(a) Oct., 1 (b) Jan. 12 (c) May 29 (d) June 12.
2. The first Africa country to host FIFA world cup is
(a) Nigeria (b) Egypt (c) Morocco (d) South Africa
3. How many members make up house of Representative in Nigeria?
(a)270 (b) 109 (c) 360 (d) 35.9
4. How many members make up senate in the upper arm of the house of assembly?
(a) 100 (b) 108 (c) 109 (d) 110
5. Into how many geopolitical zone is Nigeria divided?
(a) 4 (b)5 (c) 6 (d) 7
6. After recapitalization of banks in Nigeria, the number of banks become
(a)21 (b) 22 (c) 25 (d) 24
7. Abuja became Nigeria Federal capital territory
(a) 1991 (b) 1990 (c) 1989 (d) 1988
8. The motion for self governance was moved in Nigeria by
(a) Chief Anthony Enahoro (b) Dr. Nnamdi Azikwe
(c) Chief Obafemi Awolowo (d) Alhaji Tafawa Balewa.
9. The newest state! country in Africa is

(a) Malawi (b) South Africa (c) Southern Sudan (d) Sharawa Republic

10. When was Rosen Mubarak of EGYPT removed from office as president?
(a) Jan 2011 (b) Feb. 2011 (c) March, 2011 (d) April, 2011

II. The first civilian president that died in office in Nigeria is
(a) Sir Tafawa Balewa (b) Gen. Agunyi Ironsi
(c) Gen Murtala Mohammed (d) Alhaji Umar Yar'adua

12. Millennium development goals was grouped into how many points?
(a) 6 (b)7 (c) 8 (d) 9

13. Who was the first Military head of state in Nigeria?
(a)Gen. Olusegun Obasanjo (b) Gen Murtala Mohammed
(c) Gen. Agunyi Ironsi (d) Gen Ibrahim Babangida

14. Umar Musa Yar'adua governance was anchored on
(a) 8 point agenda (b) 7 points agenda (c) 6 point agenda (d) 10 point agenda

15. How many local government is in Nigeria?
(a) 774 (b) 744 (c) 784 (d) 794

16 Who is the First female president in Africa?
(a) Hon. Olubunmi Ette (b) Chief (Mrs.) Funmilayo Kuti
(c) Dr.(Mrs.) Ngozi Ewenla (d) Mrs. Ellen Johnson Sirleaf

17. Nigeria flag was designed by
(a) Mr. Ama Onabolu (b) Prof Wole Soyinka
(c) Prof Chinua Achebe (d) Mr. Taiwo Akinkunmi.

18. 2010 CAF African footballer of the is awarded to
(a) Samuel E'to of Cameroun (b) Dj Drogba of Cote D'Ivoire
(c) Mikel Obi of Nigeria (d) Gyan of Ghana.

19. What is the currency of India?
(a) Rupees (b) Dollar (c) Pounds sterling (c) Cfan (d) Naira.

20. Coal is mined in ……………in Nigeria.
(a) Jos (b) Enugu (c) Oloibiri (d) Igbeti.

ANSWER KEY

1.C	6 C	11 D	16 D
2.D	7 A	12 C	17 D
3.C	8 A	13 C	18 A
4.C	9 C	14 B	19 A
5.C	10.D	15 A	20 B

SECTION B

1. What natural product has been used as means of exchange long time?
(a) money (b) Dollar (c) Salt (d) Cowry

2. How many of the earth surface in percentage is covered with water?
(a) 10% (b) 29% (c) 50% (d) 71%

3. How much of the earth surface (including the rocks, mountain, valley and other physical features) in percentage is covered with land?

(a) 10% (b) 29% (c)50% (d) 71%

4. What is the smallest planet in our solar system called?

(a) mercury (b) mars (c) Venus (d) earth

5. Which language is spoken in Sicily?

(a) Spanish (b) English (c) Italian (d) Latin

6. What is the name of the famous mountain in Rio de Janeiro! Brazil

(a) Salt loaf mountain (b) Sugar loaf mountain (c) Sand mountain (d) marble mountain

7. What is the speed of light?

(a) 3 xl0^9m/s (b) 3 x10^8 m/s (c) 3 x10^5 m/s (d) 3 x10^6 rn/s

8. Which continent is largest on earth?

(a) Africa (b) Europe (c)Asia (d) Australia

9. How many months of the year have 31 days?

(a) 6 (b)7 (c) 8 (d) 9

10. What are people who do not eat protein from animal called?

(a) Omnivorous (b) Carnivorous (c) vegetarian (d) vegatis

11. From where is the Olympic fire sent out to the Olympic games every 4 years?

(a) Abuja (b) Greece (c) Washington D.C (d) United Kingdom

12. How many edge has a cube?

(a) 4 (b) 9 (c) 12 (d) 16

13. Which Continent has the coldest climate?

(a) Antarctica (b) Africa (c) Europe (d) Asia

14. Which mountain range separate Europe from Asia?

(a) The Kiliimanjaro (b) Everest (c) The Urais (d)Uranium

15. What does the word “EMIR” means?

(a) king (b) royalty (c) Prince in Arabic (d) Throne

16. Which channel connect the Atlantic ocean with the pacific ocean?

(a) Banana channel (b) panama channel (c) Palima channel (d) oceanic channel

17. When was telephone first invented?

(a) 1860 (b) 1861 (c) 1862 (d) 1862

18. Who invented the first telephone?

(a) Charles Darwin (b) Phillip Reis (c) Einstein (d) Michael Faraday

19. Cairo is the capital of which country.

(a) Morocco (b)Egypt (c) Tunisia (d) Libya

20. Lima is the capital of which country

(a) Peru (b) China (c) Chile (d) Mexico

21. Barometer is used to measure

(a) the atmospheric air (b) the atmospheric pressure (c) temperature (d) volume

22. At which temperature will pure water be transformed to steam

(a) 90° C (b) 100° C (c) 110° C (d) 115°C

23. What is a skyscraper?

(a) a mountain (b) a bird (c) a very high building (d) a tall tree

24. Which is the smallest continent?

(a) Australia (b)Africa (c) Artic & Arctatical (d) Africa

25. Which of the following gases is used to fill balloon?

(a) Oxygen (b) carbon IV oxide (c) Helium (d) Nitrogen

26. The election into the National assembly, already schedule to hold on the 2 April,2011 was shifted to
    (a) 4th April (b) 5th April (c) 7th April (d) 9th April
27. Dr. Goodluck Jonathan was sworn in as the acting president in Nigeria on the . . ..
    (a) 10th Feh.,20l0 (b) 11th feb.,2010 (c) 12th feb.,2010 (d) l3" Feb., 2010
28. Which country is hosting 2014 world cup?
    (a) USA b)Argentina c) Brazil (d)Germany
29. Who is the INEC chairman for the 2011 general election?
    a) Prof. Maurice Iwu b) Prof Wole Soyinka
    c) Prof. Attairu Jega d)Prof Chinua Achibe
30. Who was the senate president from June 2007 to June 2011 in Nigeria upper legislative chamber?
    (a) David Mark (b) Dimeji Bankole (c) Ike Ekeremadu (d) Alhaji Nafada

ANSWER KEY

1 C	6 B	11 B	16 B	21 B
2 D	7 B	12 C	17 B	22 B
3. B	8 C	13 A	18 B	23 C
4 A	9 B	14 C	19 B	24 A
5 C	10 D	15 C	20 A	25 C
26 D	27C	28 C	29 C	30.A

SECTION C

1. The vice-president in Nigeria between 1979-1983 is
   (a) Dr. Joseph Wayas (b) Dr. Alex Ekwueme
   (c) Alhaji bashiru Tofa (d) Chief M.K.O Abiola
2. A principle that advocate total equality of members of a society is called...
   (a)Communalism (b) egalitarianism (c) Totalitarianism(d) Oligarcy
3. The governor of old western region who died in a military coup with the visiting head of state in 1966 was
   (a) Lt. Col. B. S. Dimka (b) Col. Shittu Alao
   (c) Lt. Col. Adekunle Fajuyi (d) Col. Ibrahim Taiwo
4. The popular means of transportation during the trans-Sahara trade was the
   (a) donkey (b) horse (c) camel (d) mule
5. Which was the general purpose currency in pre-colonial Nigeria?
   (a) cloth (b) salt (c) copper (d) cowry
6. General Murtala Muhammad was assassinated in a coup led by
   (a) Lt. Col. Kaduna Nzeogwu (b) Col. Joe Garba
   (c)Lt.Col. B.S.Dimka (d) Major Gideon Orkar
7. The twenty one state structure came into being in Nigeria during the rule of
   (a) Gen. Murtala Muhammad (b) Major Gen. Agunyi Ironsi
   (c) Gen. Olusegun Obasanjo (d) Gen. Ibrahim Babangida
8. The colony of Lagos and protectorate of southern Nigeria were amalgamated to become the colony and protectorate of southern Nigeria in.. .
   (a) May 1900 (b) May 1902 (c) May 1905 (d) May 1906
9. The first African and only Nigerian to win nobel prize in Literature is
   (a) Prof.Chinua Achebe (b) Chris Okigbo

(c) Prof. Wole Soyinka (d) Prof. Akinwumi Ishola

10. How many Rivers is in Africa?
(a) 6 (b) 7 (c)8 (d) 9

11. Civil war broke out in Nigeria on
(a) Jan.,l967 (b) Feb., 1967 (c) May, 1967 (d) July, 1967

12. What is the third planet of the solar system?
(a) mercury (b) Mars (c) earth (d) venus

13. The solar system is made up of how many planets.
(a) 10 (b) 9 (c) 8 (d) 7

14. What is the capital of Gombe state in Nigeria?
(a) Goje (b) Gombe (c)Dutse (d) Damaturu

15. When was Ekiti state established?
(a) 1990 (b)1991 (c) 1995 (d) 1996

16. According to 2006 Censure, which state is the most populous in Nigeria?
(a) Oyo (b) Kano (c) Lagos (d) Kaduna

17. Where is marble mined in Oyo state?
(a) lgbeti (b) lgboho (c) Iseyin (d) Ogbomoso

18. Which of these is a tourist centre in Ogun state?
(a) Obudu cattle Ranch (b) Ikogusi water fall (c) Olumo Rock (d) Gurara water fall

19. Nigeria is boundaries in the North by which of the following countries.
(a) Cotonu (b)Ghana (c) Cameroun (d) Niger

20. The first election in Nigeria was held in the year
(a) 1914 (b) 1922 (c) 1948 (d) 1960

ANSWER KEY

1.B	6 C	11 D	16 B
2.B	7 A	12 C	17 A
3.C	8 D	13 B	18 C
4.C	9 C	14 B	19 D
5.D	l0.B	15. D	20 B

SECTION D

1. River Niger takes its source from
(a) mountain Everest (b) Futa Jalon (c) Kilimanjaro (d) Olumo rock

2. From Which country does River Niger takes its source?
(a) Ghana (b) Niger (c) Cote D'Ivoire (d) Guinea

3. How many days make a leap year?
(a) 362 (b)364 (c) 365 (d) 366

4. How many hours does it take for the earth to rotate on its own axis?
(a) 7 hr (b) 12 Hr (c) 24 hr (d) 36 hr

5. The last eclipse of the sun in Nigeria was observed on
(a) 29th march.2006 (b) 28th march, 2006 (c) 30th march, 2006 (d) 271h march, 2006

6. Dr. Goodluck Jonathan was sworn in as executive president on . . . .

(a) 1 May, 2010 (b) 2 May, 2010 (c) 5th May, 2010 (d) 6th May, 2010

7. Alhaji Umar Yar'adua died in office on....

(a) 1st May, 2010 (b) 2nd May, 2010 (c) 5th May, 2010 (d) 6th May, 2010

8. The first country to witness Oyster in North- Africa in 2011 is

(a) Egypt (b) Tunisia (c) Libya (d) Morocco

9. The first multi-storey building in Nigeria was built in

(a) Lagos (b) Abuja (c) Kano (d) Ibadan

10. The Premier University college was established in Nigeria in the year.

(a) 1960 (b)1947 (c) 1948 (d) 1949

11. The first television station in Africa, WNTV, was established in the year

(a) 1959 (b) 1960 (c) 1963 (d) 1966

12. Who stopped the killing of twins in Calabar?

(a) Henry Townsend (b) Mary Slessor (c) Mongo Park (d) Herbert Macaulay

13. English language Bible was translated to Yoruba Language by...

(a) Bishop Adelakun (b)Bishop Finn

(c) Bishop Ajayi Crowther (d)Cannon Sunday Makinde

14 ………. is referred to as confluence town in Nigeria.

(a) Lokoja(b) Lagos(c)Port Harcourt (d) Abuja

15. Nigeria is boundaries in the south with

(a) Pacific ocean (b) Atlantic ocean (c) Red sea (d) Artic &Antarctic

16. How many political parties participated in the 2011 general election ?

(a) 62 (b) 63 (c) 64 (d) 65

17 …….. is the leader of the first military coup in Nigeria.

(a) Col. Emeka Ojukwu (b) Major Chukwuma Nzeogwu

(c) Gen. Olusegun Obasanjo (d) Gen Yakubu Gowon

18. The basis for Nigeria's membership of the commonwealth is……….

(a) She was a former colony of Britain

(b) She was a leading opponent of Apartheid in South Africa

(c)She is the most populous black nation

(d) She provides athletes for the common wealth.

19. Which of the following is not an interest group?

(a) Christian Association of Nigeria (b) Nigerian Medical Association

(c) Catholic church (d) Pentecostal fellowship of Nigeria

20. A legislative debate or proceeding which is attended by all members of the house is called……..

(a) Plenary session (b) Recess session

(c) committee stage/ session (d) Third Reading session

ANSWER KEY

1.B	6 D	11 A	16 B
2.D	7 C	12 B	17 B
3.D	8 B	13 C	18 A
4.C	9 D	14 A	19 C
5.A	b.C	15 D	20 A

SECTION E

1. During the period of 1960-1966, Nigeria was governed under the:
(a) Presidential system of government (b) Westminster system of Government
(c) Con –federal system of government (d) Unitary system of government
2. Which of the following in the Sokoto caliphate performed functions similar to that of the Bashorun in Oyo kingdom?
(a) Waziri (b) Galadima (c) Ma'aji (d) Alkali
3. In the Igbo political system, the most senior member of the council of elder is the
(a) Okpara (b) Obi (c) Eze (d) Ofo
4. Herbert Macaulay was the first president of
(a) NCNC (a) AG (c) UMPC (d) NEPU
5. Equality before the law is a component of
(a Separation of power (b) checks and balances
(c) The rule of law (d) constitutional law
6. Adolf Hitler is to Nazism as Benito Mussolini is to
(a) Feudalism (b)communism (c) Fascism (d) Socialism
7. Nigeria became a republic on
(a) May 29, 1999 (b) Oct. 1,1960 (c) Jan. 15,1966 (d) Oct. 1, 1963
8. The Economic Community of West Africa states was established in
(a) May 1975 (b) May 1963 (c) May 1966 (d) May 1996
9. The first Indigenous Governor —General of Nigeria is
(a) Donald Cameron (b) Sir Ames Robertson
(c) Sir Adesoji Aderemi (d) Rt. Hon. Nnamdi Azikwe
10. Free-Education was introduced in western Region by which of these Premier?
(a) Chief Obafemi Awolowo (b) Chief S. L Akintola
(c) Chief Michael Adekunle Ajasin (d) Chief Bola Ige
II. The EFCC was established to
(a) arrest & try corrupt politician
(b) combat economic and financial crimes in Nigeria
(c) arrest, detain & prosecute corrupt state governors and legislators
(d) assist the world in monitoring economic project in Nigeria
12. In many countries, citizenship can be acquired through the following process except
(a) Nationalization (b) Naturalization (c) Registration (d) Birth
13. The following are Anglophone West Africa countries except
(a) Ghana (b) Nigeria (c)Kenya (d) The Gambia
14. Every political system performs the following basic function except
(a) Rule making (b) Rule transformation (c) Rule enforcement (d) Rule adjudication
15. The amalgamation of the Northern and Southern protectorates and the colony of Lagos was in
(a) 1960 (b) 1966 (c) 1963 (d) 1914
16. Globalization is all but except one of these . . . .
(a) a renewed concept in international studies (b) limited to the west
(c) a process of making the world smaller (d) an increasing integration of the world
17. The centenary anniversary of the amalgamation of Northern and Southern Nigeria will be established in
(a) 2060 (b) 2066 (c) 2064 (d)2014
18. The second military coup d'etat in Nigeria took place on

(a) Jan. 15, 1966 (b) Oct. 1, 1966 (c) July 29, 1966 (d) July 29, 1975

19. The idea of democracy stated with the

(a) Romans (b) British (c) Greeks (d) Egyptian

20. The first political party in Nigeria was established in

(a) 1923 (b) 1922 (c)l951 (d) 1979

22. Under the military regime in Nigeria, state enactment are known as

(a) laws(b) decrees (c) edicts (d) promulgation (e) proportion

23. The motto of Boys' Scout is

(a) be faithful (b) be prepared (c) be inspired (d) be serious (e) be helpful

24. Which state is referred to as 'power state' (a) Bayelsa (b) Delta (c) Niger (d) Federal capital territory (e) Kaduna.

ANSWER KEY

1 B	6 C	11 B	16 B	21 B
2 A	7 D	12 C	17 D	22 B
3. D	8 A	13 C	18 C	23 B
4 A	9 C	14 C	19 C	24 C
5 C	10A	15 D	20 B	

LADOKE AKINTOLA UNIVERSITY OF TECHNOLOGY 2006 POST UME TEST

1. A solid weighs 0.040N in air and 0.024N when fully immersed in a liquid of density 800kgm. Calculate the volume of the solid (g = 10ms2)

(a) $2.0 \times 10^{-6}m^3$ (b) $2.5 \times 10^{-6}m^3$ (c) $3.0 \times 10^{-6}m^3$ (d) $2.0 \times 10^{-6}m^3$

2. A body of mass 5kg starts from rest and is acted upon by a force 100N, the acceleration in ms^2 and final velocity after 10secs will be

(a) $20ms^2$, 20ms1 (b) $25ms^2$, 250ms (c) $20ms^2$ $200ms^4$ (d) $10ms^2$ $300ms^{-4}$

3. All these are true about impulse except

(a) change in momentum of a body (b) its unit is Ns

(c) product of force and time (d)change in acceleration of a body

4. A car of mass 100kg moves at a constant speed of 20m/s along a horizontal road where the friction force is 200N. Calculate the power developed by the engine.

(a) 2000 watt (b) 3000 watt (c) 4000 watt (d) 5000 watt

5. According to Newton's second law of motion:

(a) momentum is proportional to force action

(b) action is equal to reaction but in the opposite direction

(c) impulse force is inversely proportional to time

(d) force is proportional to rate of change of momentum

6. The weight of heat energy needed to freeze one kilogramme of milk at the inversely point is known as

(a) heat energy (b) intent heat of fusion

(c) specific latent heat of fusion (d) specific heat capacity

7. In a certain process 12,000 calories of Hexane is supplied to the system while the system does work 7500J. Find the change in internal energy: Note I cal=4.1 84J

(a) 57,708J (b) 42,708J (c) 35,208J (d) 50,208J

8. Some water is heated in a pot. The major mode(s) of the heat transfer through the water is/are by:

(a) convection (b) conduction (c) radiation (d) conduction and radiation

9. Calculate the quantity of heat required to completely convert 20kg ice at 0^0C of water at the same temperature (specific latent heat of fusion of ice = 336Jkg'
(a) 8.06KJ (b) 706KJ (c) 538KJ (d) 6.72KJ

10. An object of mass 10g requires 20J of heat energy to change its temperature by 20^0C. Calculate the specific heat capacity of the object.
(a) $0.1Jg^{-1}K^{-1}$ (b) $0.2g^{-1}K^{-1}$ (c) $0.4Jg^{-1}K^{-1}$ (d) 0

11. Which of these radiation does not originate with the nucleus?
(a) Alpha (b) x-rays (c) beta (d) neutron

12. A tyre is pumped to a pressure of 30Nm2 at 27°C, when the tyre rates up to 54°C. Find the new pressure assuming no change in volume.
(a) 40.7Nm 2 (b) 60.0Nm2 (c) 32.7Nm 2 (d) 52.6Nm 2

13. An eclipse of the sun may occur when
(a) the sun passes between the moon and the Earth
(b) the sun passes between the Earth and the sun
(c) the Earth passes between the moon and the sun
(d) the moon and the Earth rotate together

14. The shortest mirror in which a person. interest can see its entire image is
(a) 2m (b) 0.5m (c) 4m (d) 1m

15. The nature of the image formed by an object placed 12cm from a con verging lens of focal length 18cm is
(a) virtual and magnified (b) virtual and real (c) a infinity (d) virtual

16. A erect image three line the size of the object is obtained from a concave mirror of radius of curvature 36cm, what is the position of the object from the mirror?
(a) 24cm (b) 12cm (c) 10cm (d) 8cm

17. The angle of deviator of light of various colours passing through a glass prism decrease in the order
(a) red, blue and orange (b) blue, red, orange (c) red, orange, blue (d) orange, blue, red

18. Two positive point charge of 12μC and8μC respectively are 10cm apart. The work done is bringing them 4cm closer is (K= 9×10^2 N^2mc^2)
(a) 600J (b) 300J (c) 6.8J (d) 518J

19. The effective capacitance of a system of capacitors arranged such that a 4μF capacitor is in series with 3μF.
(a) 1.7μF (b) 3.0μF (c) 4F (d) 5.0μF

20. How long will it take to heat 3kg of water from 28^0C to 88^0C using an electric kettle, which taps 6A from a 210V supply?
(a) 5.6 minutes (h) 9.6 minutes (c) 10.0 minutes (d) 19.3 minutes

21. A moving cell galvanometer has a resistance of 10and a full scale deflection of 0.0 IA, it can be converted into a voltmeter of 10V full-scale deflection by connecting a resistor of
(a) 969Ω parallel (b) 990Ω parallel (c) 10Ω parallel (d) 10Ω parallel

22. A musical note is different from a noise in that the
(a) amplitude of a noise is greater than that of a musical note
(b) frequency of a musical note is regular while that of a noise is irregular
(c) wavelength of a musical note is longer than that of a noise
(d) frequency of a note is higher than that of a musical note

23. Any line or section through an advancing wave in which all the particles are in the same phase is called
(a) wave crest (b) wave tough (c) wave amplitude (d) wave front

24. Which of the following diameters is the best absorber of x-rays/gamma ?
(a) hydrogen (b) calcium (c) copper (d) lead

25. Thunder is usually heard some seconds after lighting is seen because
 (a) sound and light travel in different media
 (b) thunder occurs after lighting
 (c) sound travel more slowly than light
 (d) sound travels in the form of waves but light does not

26. If a source of sound is moving, a stationary listener will hear a sound of different frequency. This is called
 (a) Doppler effect (b) resonance (c) refraction (d) diffraction of sound

27. The period of vibration of a wave of wavelength 30m moving at a speed of300m/s
 (a) 10s (b) 270s (c) 330s (d) 900s

28. The quality (timber) of sound depends
 (a) amplitude (b) frequency (c) harmonies (d) wavelength

29. Which of the following consist entirely of scalar quantities:
 (a) pressure, work and electric potential
 (b) force, momentum and distance
 (c) velocity, energy and impulse
 (d) mass, time and temperature

ANSWER KEY

1 A	7 A	13 C	19 A	25 C
2 C	8 A	14 A	20C	26 A
3 D	9 D	15 A	21 B	27 A
4 C	10 A	16 A	22 B	28 C
5 C	11 A	17 D	23 D	29 D
6 C	12 C	18 D	24 D	

EXPLANATION TO ANSWERS

1. Weight of liquid displaced = Upthrust

 = (0.040 - 0.024)N

 = 0.016N = 0.016/10 = 1.6×10^{-3}kg

 Volume of the object = Volume of liquid displaced

$$= \frac{Mass\ of\ liquid}{density\ of\ liquid} = \frac{1.6\ x\ 10^{-1}kg}{800kgm^{-2}}$$

 = $2.0 \times 10^{6}m^{3}$ (A)

2. Ft = mv - mu

 Ft = m (v-u)

 100 x 10 = 5(v-0)

 1000 = 5v

 V = 200m/s

 V = u – at

200 = 0 – mass

a = 20m/s^2 (C)

3. Impulse, Ft = m(v-u)

➔ $F = \frac{m(v-u)}{t}$ (D)

4. Power, P $= \frac{work\ x\ distance}{time} = Fy$

Power P = FV = (200N (20m/s)

= 4000W (C) V = velocity

5. The second law of motion states that a body which is subjected to a net
external force (F) acquires
acceleration which is directly proportional to the force and inversely proportional to the mass of the body i.e. F = ma (C)

6. Specific latent heat of fusion (C)

7. Since I calorie = 4.184J

12,000 calorie = (4.184 x 12,000)J = 50.208Jheat supplied = Work done

$(\Delta W) + \Delta U\ (Internal\ Energy)$

50,208 = 7500J + ΔU

ΔU = (50.208 – 7500)J

$= 42708J$ (A)

8. Convection is the transfer of heat from a hotter region of a Fluid to a colder region through the actual motion of the fluid (A)

9. Heat required to change the phase of

mass, in of the substance is Q = rnL$_f$

= 20kg x 336Kg^{-1}

= 6.72 x 10^{3}J (D)

= 6.72kJ

10. Specific heat capacity = m x c x $\Delta\theta$

= (0)(20)(20)

Required, H = mcθ

20 = 10 x c x 20K

$c = \frac{20}{200} = 0.1Jg^{-}K^{-1}$ (A)

11. Apha decay produces Helium particle

2He (A)

12. $\frac{Pt}{T2} = \frac{P2}{T2}$

$\frac{30}{(27+273)} = \frac{P2}{(54+273)} = \frac{30}{300} = \frac{P2}{327}$

$P2 = \frac{(30)(327)}{300} = 32.7Nm^{-2}$ (C)

13. When the moon comes between the sun and the earth, the shadow of the moon may fall on the Earth. If one is living on the part of the Earth that is in moon's shadow, the light from the sun cut off and we are in darkness. Since we cannot see the sun we call this an eclipse of the sun (C)

14. A

15. f = 18cm & object distance, u = 12

$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$

$\frac{1}{12} + \frac{1}{v} = \frac{1}{18}$

$\frac{1}{v} = \frac{1}{18} - \frac{1}{12}$

$= \frac{1}{v} = \frac{2-3}{36}$

$\frac{1}{v} = \frac{1}{36}$

V = -36cm

16. Object distance, U = 24cm (A)

17. Red light has the highest speed and it is the least deviated while violet with the lowest speed is the most deviated (D)

18. Work done Force x Distance

Force F = $\frac{kq2q2}{r^2}$

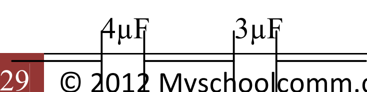
K = 9 x 10^9NM^{2}c^{-2} q1 = 12μC

Q_2 = 8 μC, r = 10cm d = 6cm

F = $\frac{9 \, x \, 10^9 \, x \, 12 \, x \, 10^{-6} \, x \, 8 \, x \, 10^{-6}}{(0.1)^2} = 86.4\mu$

Work done = 86.4 x 6

= 518.4J (D)



19.

$\frac{1}{CE} = \frac{1}{C1} + \frac{1}{C2}$

$\frac{1}{4} + 3 = \frac{3-4}{12}$

$\frac{1}{CE} = \frac{7}{2}$

CE $= \frac{12}{7} = 1.7\mu\text{F}$ (A)

20. Heat loss by kettle = heat gained by water

1vt = mcθ

(6) (210)t = (3)(60)(4200)

1260t = 756000

t = 60secs = 10min (C)

21. Since the total current which can pass through the galvanometer is O.D.I.A. The total p.d. across the galvanometer and the resistanceRis0.OIA(10+R)=10

10+Rz 1000

R=1000—10=990Ωinseries (B)

22. B

23. Wave front is any line or section taken through an advancing wave in which all the particles are in the same phase (D)

24. Gamma-Rays have the highest penetrating capacity. It can only be stopped using leadblock (D)

25. Sound travels at lower speed compared to light (C)

26. A variation in the perceived sound frequency due to the motion of the sound source is an example of the Doppler effect (A)

27. $f = \frac{v}{\lambda} = \frac{300m/s}{30m} = 10^{-1}$

$frequency = \frac{1}{t} = \frac{1}{10secs} = 0.1secs \quad (No\ option)$

28. The harmonics present in a given note determine the quality of a note (C)

29. Mass, time and temperature are all scala rquantities. They possess magnitude but no direction (D)

LADOKE AKINTOLA UNIVERSITY OF TIC1INOLOGY (2007) POST UME TEST

1. A car accelerates from $3rns^{-1}$ to 100ms in 35s. Calculate the acceleration of the car.
   (a) $2ms-^{2}$ (b) 3ms2 (c) $4ms^{2}$(d) $5ms^{-2}$
2. A body bf mass 2kg moves round a circle of' radius 2m with a constant speed of 165m/s. Calculate the force toward the centre.
   (a) 40N (h) 50N (c) 80N (d) 100N
3. A girl of mass 65kg falls horn a bridge that 50m above the water. Calculate her kinetic energy after falling for 5 sees.
   (a) 1 125J (b) 28125J (c) 9375J (d) 812501
4. Which of the following is a mechanical wave?
   (a) micro wave (b) water wave (c) X-ray (d) infra-red ray
5. The half-life of the radioactive substance of mass 100g is 30 minutes. What fractions of the original mass is left over after 3hrs?
   (a) 3.125 (b) 6.25 (e) 1.5625 (d) 0.02563
6. Two resistors 12Ω and 8Ω are connected in parallel, if the power developed in 12Ω, resistor is 6w, determine the power developed in 3.32 resistor.
   (a) 9w (b) 12w (c) 16w (d) 24w
7. How long will a 60 wart element take to heat water of mass 0.5kg at $30^{0}C$ to $90^{0}C$. Neglect heat loss (S.H. C. of water = $4200jkg^{-1}K^{-1}$)
   (a) 350secs (b) 1260secs (c) 2100secs (d) 1500secs
8. The critical angle of glass is 42°, the approximate refractive index of this glass is
   (a) 1.50 (b) 1.38 (c) 1.80 (d) 2.30

EXPLANATION TO ANSWERS

1. Final velocity, V = 100m/s
   Initial velocity u = 3m/s
   V = U + at
   Q= (100 – 3) / 35 =97 / 35 =$2.77m/s^{2}$
   ≈ $3.0m/s^{2}$ (B)

2. Centripetal force = mv^{2}/r
   $= \frac{2*(63)^2}{2}$
   =40N (A)

3. v= u + at
   S= ut $+1/2at^{2}$
   50 = 0 + ½ x a x $(5)^{2}$
   250 = 25a
   a = $10m/s^{2}$

$v = u + at$

$v = 0 + 10 \times 5 = 50m/s$

$K.E = 1/2mv^2$

$= ½ \times 65 \times (50)^2$

K.E. = 81250J (D)

9. Second harmonic motion of a string 1.5m long fixed at both ends 80Hz. The speed of transverse wave in the string is

(a) 120ms (b) 120m/s (c) 2.12m/s

(d) 102m/s

10. The echo produced when a motion is fire from a ship some distance from cliff is heard 1.2 second. Calculate the distance of the ship (if the velocity of Sound in air = 330m/s)

(a) 138m (b) 198m (c) 275m (d) 296m

ANSWERS KEY

1.B	2.A	3.D	4.B	5.C
6.A	7.C	8.A	9.B	10.D

4. Mechanical wave travels through a material medium. Examples are sound, water and water wave. (13)

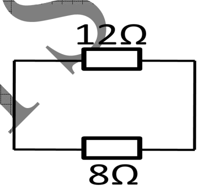
5. Original mass,

$100g \xrightarrow{30\ mins} 50g \xrightarrow{30\ mins} 25g \xrightarrow{30\ mins} 12.5g \xrightarrow{30\ mins}$

$6.25g \xrightarrow{30\ mins} 3.125g \xrightarrow{30\ mins} 1.5625g$

After 3hrs, 1.5625 is left (C)

6



$\frac{1}{R} = \frac{1}{12} + \frac{1}{8}$

$\frac{1}{R} = \frac{2+3}{24}$

$\frac{1}{R} = \frac{5}{24}$, $R = \frac{24}{5}$

7 Heat loss by element = that gained by water

$60 \times t = 0.5 \times 4200 \times 60$

LADOKE AKINTOLA UNIVERSITY

TECHNOLOGY 2008 POST UME TEST

1. To what height will a missile find vertically upwards with a speed of 28m/s attain if air resistance is neglected? g = 9.8m/s

2. Which of the following is not an example of a machine?

(a) screw (b) lever (c) horizontal plane (d) pulley

3. How long will it take to heat 3kg of L1't from 28°C in an electric kettle taking 6A from 220V supply? (specific heat capacity of water is 4180J/k)

(a) 2mins (b) 9mins (c) 1Omins

(d) 20mins

4. A cell with an emf of 1.5V and an internal resistance of 10.Oohms is connected to two resistances of 2.0 and 3.0 ohms in series. The current produced by this cell is

(a) 0.25A (b) 2.5A (c) 0.5A (d) 0.75A

5. Which of the following is not essential for the production of electron by thermionic emission?

(a) Tungsten target (b) Reflection plate (c) Fluorescent (d) Hot cathode

6. If an object moves with a constant speed round a circle, it has an acceleration which is

(a) constant in magnitude and varying in direction

(b) varying in magnitude and constant in direction

(c) constant in magnitude and direction

EXPLANATION TO ANSWERS

1. H = 1/2gt

$V^2 = u^2 + 2as$

$28^2 = 0 + 2(9.8)S$

S = 40m (B)

2.- A simple machine is a device which enables mechanical work to be performed in a convenient way. An horizontal plane can not serve this purpose (C)

(d) varying in magnitude and direction

3 Heat gain = Heat loss

$Ivt = mc\Delta\theta$

$6 \times 220 \times t = 3 \times 4180 \times 57$

T = 541.1secs = 9.025mins (B)

4 V = I(R + r) = $\frac{1.5}{(5+1)}$ = $\frac{1}{4}$

= 0.25A (A)

7. An electric string of length X is stretched through a length by force F. The area of cross-section of the string is A and its young modulus is E.

Which of the following expression is correct?

(a) F= $\frac{EAe}{X}$ (b) F = $\frac{EAe}{X^2}$ (c) F = $\frac{EAe^2}{X}$ (d) F = $\frac{EAe^2}{X^2}$

8. The density of a solid is 130gcm3 at the temperature of 25°C. Find the density at 150°C if the linear expansion of the solid is 2.0 x 105K1.

(a) 21.90gcm3 (b) 9.021.90gcm3 (c)12.9gcm3 (d) 20.521.90gcm3

9. A bullet leaves the barrel of a gun with a speed of 80ms'. If the barrel is 20m long. Find the acceleration of the bullet.

(a) l6Oms' (b) 40ms' (c) 800ms' (d) 1600ms1

10. Sound and water waves are classified as mechanical waves and they cannot be propagated in one of the following media

(a) air (b) water (c) steel
(d) vaccuum

5 Thermionic emission is when a piece of metal placed in a vacuum is heated and electrons are emitted from the metal surface. Tungsten is a metal and can emit electrons when heated.

6 A body which moves in a circular path of radius r, moves at a constant speed (of magnitude, v) while its direction of motion changes continuously as it moves round the circular path (A)

7. The Young's modulus of elasticity is the ratio of tensile stress strain within the elastic region. i.e. Young modulus,

$$E = \frac{\text{Tensile stress}}{\text{Tensile strain}}$$

$$\text{Tensile stress} = \frac{\text{tensile force}}{\text{Cross–sectional area}}$$

$$= \frac{F}{A}$$

Tensile stress = Extension = e

Original x

$$E = \frac{F/A}{e/x} = \frac{FX}{Ae}$$

$\Rightarrow$ $F = \frac{EAe}{X}$ (A)

8. linear expansitivity, a = 2.0 x 10^{-5}k
Cubic expansitivity, y = 3a
$3(2x10^{-5}) = \frac{\Delta v}{V1 \times 125}$

$\frac{\Delta v}{v_1} = 750 \times 10^{-5}$

Since is constant; $m^1 = m^2$

$$\Rightarrow D_1 = \frac{m_1}{v_1} \& D_{1=} \frac{m_2}{v_2}$$

$D_1V_1 = D_2V_1 \& V_2 = V_1\ \Delta V$

$$\& \Delta V = 750\ x\ 10^{-5}V_1$$

$$\Rightarrow 13.0\ x\ V_1 = D_2(V_1 + \Delta V)$$

$$13.0\ x\ V_1 = D_2(V_1 + 750\ x10^{-5}V_1)$$

$$13.0V_1 = D_2(1 + 750\ x10^{-5})V_1$$

$$13.0V_1 = (1 + 750\ x10^{-5}V_1)D_2$$

$$D_2 = \frac{13.0}{1.0075} = 12.90cm^3(C)$$

9. velocity, V = 80m/s
Length of barrel = 20m

$$V^2 = U^2 + 2as$$

$$(80)^2 = 0^2 + 2\ x\ a\ x\ 20$$

6400 = 40a
A= 160m/s^2 (A)

10. Mechanical waves need a material medium to travel, sound and water wave cannot travel through a vacuum. (D)

LADOKE AKINTOLA UNIVERSITY

2009 POST UME TEST

1. A 5kg at rest is acted upon by a force of 20N for 20 milliseconds. The increase in momentum final speed of the body respectively are
   a) 4Ns and 8Ns
   b) 0.4Ns and 0.08Ns
   c) 0.4Ns and 0.8Ns
   d) 4Ns and 0.8Ns
2. A body throws a stone up to a height of 10m and catches it back. What is the displacement of the stone (assume the hand remains at the same level at throw and at catch)
   a) 10m (b) 20m (c) 30m d) 0m
3. A body moves along a circular path with uniform angular speed of 0.6rads and at a constant speed of 3.0m/s. Calculate the acceleration of the body towards the centre of the circle.
   a) 2.5m/s b) 5.4m/s b) 5.4m/s c) 5.0m/s d)1.8m/s
4. Which of the following is a derived unit?
   a) Newton (b) Kelvin c) Kilogram (d) second
5. The change of the direction of a wave as a result of a change in the velocity of the wave in another medium is called.
   a) Refraction
   b) Diffraction
   c) Interference
   d) Polarization
6. Light of frequency 6.0×10^{14}Hz travelling in air is transmitted through glass of refractive index 1.5. Calculate the frequency of the light in the glass.
   a) 6.0×10^{14}Hz b) 4.0×10^{14}Hz
   b) 7.5×10^{14}Hz d) 9.0×10^{14}Hz
7. A 70^0 glass prism has a refractive index of 1.5. Calculate the angle of incidence for minimum deviation
   a) 35^0 b) 49^0 c) 59^0 d) 45^0
8. An object placed 12cm in front of a convex lens produces a virtual image of magnification 3.0. the focal length of the lens is
   a) 9cm b) 12cm c) 18cm d) 36cm
9. What is the average velocity of the molecules in a sample of oxygen at 100^0C? the mass of oxygen is 5.3×10^{-26}kg?
   a) 540m/s b) 450m/s c) 540m/s
   d) 504m/s
10. A conductor has a diameter of 1.0mm and length 2.0mm, if the resistance of the material is
   a) $2.55 \times 10^5 \Omega$m
   b) $2.55 \times 10^2 \Omega$m
   c) $3.93 \times 10^{-5} \Omega$m
   d) $3.93 \times 10^{-8} \Omega$m

LADOKE AKINTOLA UNIVERSITY
2010 POST UME TEST
MATHEMATICS

1. Solve for X if $4^{3x-2} = 26^{x=1}$
   (a) 6,694 (b) 6,649 (c) 6,469 (d) 6,496
2. A cylinder of base radius 4cm has a volume of $100cm^3$. Calculate its height.
   (a) 1.99cm (b) 9.19cm (c) 9.91cm (d) 10cm
3. Find the real value of x for which sin hx = 1.475
   (a) 1.8108 (b) 1.0881 (c) 1.1808 (d) 1.1081
4. If $4\pi r^3 / 3 = 128.1$, then what is $4\pi r^2$?
   (a) 132 (b) 123 (c) 213 (d) 321
5. Solve the polynomial
   $x^3 + 4x^2 + x - 6 = 0$
   (a) 0, -1, 2 (b) -2, 4, 5 (c) 1, -3, -4 (d) 0, 1, 2
6. The sides of a Rhombus are 4.2cm each. One of its angle is 58^0. Calculate the lengths of the diagonals.
   (a) 4.07, 7.34 (b) 4.70, 7.34 (c) 4.34, 7.07 (d) 4.70, 7.43
7. Calculate the mean, median and mode of 2, 3, 3, 3, 5.
   (a) 3,2,3,2 (b) 3,2,3,3 (c) 2,3,3,3 (d) 3,2,2,3
8. Solve for x in the equations 6x + 4y = 5 and 3x + 2y = 5
   (a) x = 0(b) x = 2(c) x = 5 (d) No solution
9. Let $y = \log a^x$, calculate dy / dx
   (a) 1 / asinx (b) $1 / x^2 \ln a$ (c) 1/2x In a (d) 1 / xlna
10. Evaluate $f(2)$ if $f(x) = \frac{\sin 45x}{x^2 - 2}$
    (a) 2 (b) ½ (c) undefined (d) 0

SOLUTION

1. $4^{3x-2} = 26^{x=1}$
   Taking the logarithm of both sides
   (3x – 2) log4 = (x + 1) log26
   3xlog4 - 2log4 = x log 26 + log 26.
   3x log 4 – x log 26 = log 26 + 2log 4
   x (3 log4 log 26) = log26 + 2log 4
   dividing both sides by 3 log4 – log26
   $x = \frac{\log 26 + 2\log 4}{3\log 4 - \log 26}$

   $= \frac{1.4150 + 2(0.6021)}{3(0.6021) - 1.4150}$

   $= \underline{2.6192}$

0.3913 = 6.6914 (A)

2. The volume (V) of a cylinder with base radius (r) and height (h) is given by $V = \pi^2 h$

$V = 100cm^3$, r = 4cm, h = 7

$\Rightarrow 100 = \pi \times 4^2 \times h$

$100 = 22/7 \times 16 \times h$

$\Rightarrow h = \dfrac{10 \times 7}{22 \times 16}$

= 1.99cm (A)

3. Recall that, $\sin hx = \dfrac{e^x - e^{-x}}{2}$

$\therefore \dfrac{e^x - e^{-x}}{2} = 1.475$

Multiply both sides by 2, we have

$e^x - e^{-x} = 2.95$

Multiplying through by e^x

$e^{2x} - 1 = 2.95e^x$

$\Rightarrow e^{2x} - 2.95^x - 1 = 0$

Put $e^x = p$ .... (*)

$\Rightarrow p^2 - 2.95p - 1 = 0$

Using general formula for solving quadratic equation

$P = \dfrac{2.95 \pm \sqrt{(-22.95)^2 - 4 \times 1 \times (-1)}}{2}$

$= \dfrac{2.95 \pm 3.5641}{2}$

$\Rightarrow p = 3.2571$ or -0.3071

when p = -0.3071 we have from (*) that

$e^x = -0.3071$. This is not possible since exponential function is non negative.

When p = 3.2571, we have from (*) that

$e^x = 3.2571$

$\Rightarrow x = \log_e (3.2571)$

$= \ln 3.2571$

$= 1.1808$ (C)

4. $4\pi r^3 / 3 = 128.1$

Dividing both sides by $4\pi / 3$

$r^3 = \dfrac{3 \times 128.1}{4\pi}$

$\Rightarrow r = \sqrt[3]{\dfrac{3 \times 128.1}{4\pi}}$

$= \sqrt[3]{\dfrac{3 \times 128.1 \times 7}{4 \times 22}}$

$= 3.13$

$\therefore\ 4\pi r^2 = 4 \times 22/7 \times (3.13)^2$

≈ 123 (B)

5. $x^3 + 4x^2 + x - 6 = 0$

Let $P(x) = x^3 + 4x^2 + x - 6$

Now, $P(1) = 1 + 4 + 1 - 6 = 0$. Hence, $x-1$ is a factor of $P(x)$ by factor theorem.

$$
\begin{array}{r|l}
 & x^2 + 5x + 6 \\
\cline{2-2}
x - 1 & x^3 + 4x^2 + x - 6 \\
 & x^3 - x^2 \\
\cline{2-2}
 & \qquad 5x^2 + x \\
 & \qquad 5x^2 + x \\
\cline{2-2}
 & \qquad\qquad 6x - 6 \\
 & \qquad\qquad 6x - 6 \\
\cline{2-2}
 & \qquad\qquad\quad 0
\end{array}
$$

$X^2 + 5x + 6 = x^2 + 2x + 3x + 6$

$= (x+2+(x+3)$

$\therefore\ x^3 + 4x^2 + x - 6 = 0$

$\Rightarrow (x-1)(x+2)(x+3) = 0$

$\Rightarrow$ either $x - 1 = 0$ or $x + 2 = 0$

or $x + = 0$

$\therefore\ x = -3, -2, 1$

1^{st} diagram

If $ABC = 58^0$, then $BCD = 180^0 - 58^0 = 122^0$

From Δ BCD

$c^2 = b^2 + d^2 - 2bd \cos C$ (cosine rule)

$= 42^2 + 4.2^2 - 2 \times 4.2 \times 4.2 \cos 122^0$

$c = \sqrt{35.28 + 18.7}$

$= 7.34$cm

From Δ BCD

$b^2 = a^2 + c^2 - 2ac \cos B$ (cosine rule)

$b^2 = 42^2 + 4.2^2 - 2 \times 4.2 \times 4.2 \times \cos 58^0$

$b = \sqrt{35.28 - 18.7}$

$= \sqrt{16.58}$

$= 4.07$cm

$\therefore$ The lengths of the diagonals are 4.07cm and 7.34cm (A)

7. Mean $= \dfrac{2+3+3+3+5}{5} = 3.2$

Median = 3;

Mode = 3 (B)

8. $6x + 4y = 5$ …………….. (i)

$3x + 2y = 5$ …………….. (ii)

Equations (i) and (ii) are parallel lines since

$6x + 4y = 2(3 + 2y)$.

Therefore, there is no solution (D)

9. $y = \log a^x$

$\Rightarrow y = \dfrac{\log a^x}{\log e^a}$

$\Rightarrow y \log e^a = \log a^x$

Differentiating through with respect to x

$\text{Log}e^a \, dy / dx = 1/x$

$\Rightarrow dy/dx = 1/x\log e^a = 1/x\ln a$ (D)

10. $f(x) = \dfrac{\sin 45^0 x}{X^2 - 2}$

$= \dfrac{\sin 90^0}{4 - 2} = \dfrac{1}{2}$ (B)

LADOKE AKINTOLA UNIVERSITY
2006 POST UME TEST
MATHEMATICS

1. If two graphs $y = px^2 + q$ and $y = 2x^2 - 1$ intersect at $x = 2$, find the value of p in terms of q.

(a) $\dfrac{7-q}{4}$ (b) $\dfrac{q-8}{7}$ (c) $\dfrac{8-q}{2}$

(d) $\dfrac{7+q}{8}$

2.

Interval years	10-12	13-15	16-18	19-20	21-23
No. of pupils	6	14	15	10	5

The table above shows the frequency distribution of the ages (years) of pupils in a certain Sunday School. What percentage of the total number of pupils is over 15 years but less than 21 years?

(a) 60% (b) 50% (c) 45%

(d) 35%

3. Evaluate $\begin{vmatrix} -1 & -1 & -1 \\ 3 & 1 & -1 \end{vmatrix}$

1 2 1

(a) -12 (b) 4 (c) -4 (d) -2

4. Given the matrix K = 2, the matrix $K^2 + K + 1$ when 1 is 2 x 2, identify matrix is

(a) 6 3 (b) 7 2 (c) 9 8 (d) 10 7

5. If y/2 = x, evaluate $x^3/y^2 + ½ + (½ - x^2/y^2)$

(a) -5/2 (b) 5/8 (c) 5/16 (d) 5/2

6. Let p = (1,2,u,v,w,x),

Q = (2,3,u,v,w,5,6,y) and

R = 2,3,4,v,x,y),

Determine (P/Q) $\cap$ R

(a) ϕ (b) (x) (c) (x,y) (d) (1,x)

7. If y = 2xcos 2x – sin 2x, find dy/dx when x = π

(a) π (b) 0 (c) 2π (d) π

8. A binary operation * is defined by $a*b = a^b$ if a*2=2 – a, find possible value of a.

(a) -1,2 (b) -2,7 (c) 1,2 (d) 1,2

9. If a and β are the toots of the equation $3x^2 – 5x^2 – 2 = 0$, find the value of 1/a + 1/b

(a) 5/2 (b) -5/2 (c) (d) -1/2

10. The 3rd term of an AP is 4x – 2y and the 9th term is 10x – 8y, find the common difference.

(a) x,y (b) 19x-17y (c) 2x (d) 8x-4y

11. Convert 38 in base 10 to a number of base 2.

(a) 1001 (b) 11011 (c) 100101

(d) 111001

12. Find the curved surface area of a cone whose base radius is 6cm and whose height is 8cm

(a)$18\pi cm^2$ (b) $60\pi cm^2$

(c) $24\pi cm^2$ (d) $56\pi cm^2$

13. If log $32(x^2 + 2x + 3) = 1/5$, find the value of x

(a) 2 or 3 (b) -1twice (c) 1 or -2

(d) 1 twice

14. Rationalize the denominator $6 + 2\sqrt{5}/_4 - 3\sqrt{6}$

(a) $\frac{12+4-5+7x\ 6+3\ x\ 30}{19}$

(b) $\frac{-(15+3x5 + 15\sqrt{6} + 7\sqrt{30})}{3}$

(c) $\frac{-(24 + 8 - 18\sqrt{6} + 6\sqrt{30})}{39}$

(d-) $\frac{-(12+4\sqrt{5} + 9\sqrt{6} + 3\sqrt{30}}{19}$

15. A ladder resting on a vertical wall makes an angle whose tangent is 2.4 and the distance between the foot of the ladder and the wall is 50cm. what is the length of the ladder?

(a) 1m (b) 1.1m (c) 1.3m (d) 1.2m

16. Find the value of x and y in the equations $2x + 5y = 11$, $7x + 4y = 2$

(a) x=8, y=0 (b) x=2, y=-3

(c) x=34/27, y=73/27

(d) x = -27/34, y=73/27

17. If $y = 2x^2 + 9x – 35$, find the ranges of values of which y ≤ =

(a) $-5 \le x \le 7$ (b) $-7 \le x \le 5/2$

(c) $-7/2 \le x \le 5$ (d) $-5 \le x \le 7$

18. If the first term and the sixth term of the geometric progression is 9 and 32/27, calculate the sum of the first four terms?

(a) 212/3 (b) 23 (c) 23½ (d) $36^2/_3$

19. Sin θ = $\sqrt{3}/2$ are less than 90^0.

$\tan \dfrac{(90 - \theta)}{\cos^2\theta}$

(a) x3/2 (b) ½ (c) 4/√3 (d) 2√3

SOLUTION

1. Hint: At the point(s) of intersection of the graph $y = f_1(x) = f_2(x)$. Therefore, at the 6 point of intersection of $y = px^2 + q$ and $y = 2x^2 - 1$

$px^2 + q = 2x^2$

$q + 1 = 2x^2 – px^2$

$q + 1 = x^2 (2 – p)$ by factorizing x^2

Dividing both sides by 2 – p

$X^2 = \dfrac{q + 1}{2 – p}$

Substituting x = 2 (the given point of intersection)

$2^2 = \dfrac{q + 1}{2 – p}$

$4 = \dfrac{q + 1}{2 – p}$

Multiplying both sides by 2 – p

$8 – 4p = q + 1$

$=> = \dfrac{7-q}{4}$ (A)

2. Total number of pupils in the Sunday school = 50

Total number of pupils above 15 years but less than 21 years = 15+10 = 25

Percentage of the total number of pupils above 15 years but less than 21 year = $^{25}/_{50}$ x 100% = 50% (B)

3. $\begin{vmatrix} -1 & -1 & -1 \\ 3 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix}$

= $-1\begin{vmatrix}1 & -1\\ 2 & 1\end{vmatrix} -(-1)\begin{vmatrix}3 & -1\\ 1 & 1\end{vmatrix} +(-1)\begin{vmatrix}3 & 1\\ 1 & 2\end{vmatrix}$

= -1(1+2) + 1(3+1) - 1(6-1)

= -34 – 5 = -4 (C)

4.

5. To evaluate $(x^2/y^2 + ½)+(½ - x^2/y^2)$

Given that y/2 = x

y/2 = x => x/y = ½

Hence, $(x^2/y^2 + ½)+(½ - x^2/y^2)$

= $([x/y]^3 + ½) + ½ - [x/y]^2)$

Putting x/y = ½

= $([½]^3 + ½) + (½ - [½]^2)$

= (1/8 + ½) + (½ - ¼)

= 5/8 + ¼

5/8 x 4/1 = 5/2 (D)

6. P/Q = (1,x)

.: (P/Q) $\cap$ R = (x) (B)

7. y = 2x cos 2x – sin 2x

dy/dx = d/dx (2x cos 2x)–d/dx sin2x

= 2 cos 2x – 4xsin2x – 2cos 2x

= -4x sin 2x

.: when x = π, dy/dx = -4 x π x sin2π

= -4π x 0 (sin 2π = 0)

= 0 (B)

8. a * 2 = 2 - a

=> a^2 = 2 – a

=> a^2 + a – 2 = 0

=> a^2 + 2a – a – 2 = 0

=> a(a+2) – 1(a+2) = 0

=> (a+2) (a-1) = 0

=> a+2=0 or a-1 = 0

=> a = -2 or 1 (B)

9. Given α and β as the roots of the equation $3x^2 – 5x – 2 = 0$

The $\alpha + \beta$ = 5/3; $\alpha\beta$ = -2/3

.: $1/\alpha + 1/\beta = {}^{\beta+\alpha}/_{\alpha\beta}$

= 5/3 + -2/3 = 5/3 x -3/2

= 5/2 (B)

10. nth term of an AP. With first term 'a' and common different 'd' = a+(n-1)d

3^{rd} term = 4x – 2y and 9^{th} term = 10x -8y

=> a + 2d = 4x = 2y …… (i)

a + 8d = 10x – 8y …………(ii)

subtract (i) from (ii)

6d = 6x – 6y

Dividing through by 6, we get

d = x-y (A)

11.

2	38
2	19 r 0
2	9 r 1
2	4 r 1
2	2 r 0
2	0 r 1

.: $38_{10} = 100110_2$ (A)

12. Curved surface area of a cone = πrl

2^{nd} Diagram

$L = \sqrt{h^2 + r^2}$

.: C.S.A. = $\pi \times 6 \sqrt{8^2 + 6^2}$

= $\pi\, 6 \times 10 = 60\pi cm^2$ (B)

13. $\log_{32} (x^2 – 2x + 3) = 1/5$

=> $x^2 – 2x + 3 = 32^{1/5}$

$X^2 – 2x + 3 = 2$

=> $x^2 – 2x + 1 = 0$

=> $(x – 1)^2 = 0$

=> x = 1 twice

14. $\frac{6 + 2\sqrt{5}}{4 - 3\sqrt{6}}$

Rationalize

$\frac{6 + 2\sqrt{5}}{4 - 3\sqrt{6}} \times \frac{4 + 3\sqrt{6}}{4 - 3\sqrt{6}}$

$\underline{24 + 18 \sqrt{6} + 8\sqrt{5} + 6\sqrt{30}}$

= $16 + 12\sqrt{6} - 12\sqrt{6} - 9\,(6)$

= $\dfrac{24 + 18\sqrt{6} + 8\sqrt{5} + 6\sqrt{30}}{16 - 54}$

= $\dfrac{2(12 + 9\sqrt{6} + 4\sqrt{5} + 3\sqrt{30}}{-38}$

= $\dfrac{-(12 + 9\sqrt{6} + 4\sqrt{5} + 3\sqrt{30}}{9}$ (D)

15. **Diagram 3**

The length of the ladder is /AB/

Using trig. ratio

$\tan\theta = \dfrac{/AC/}{50}$

=> $/AC/ = 50 \times \tan\theta$

$= 50 \times 2.4 = 12$cm

Using Pythagoras theorem

$/AB/ = \sqrt{/AC/^2 + /B/C/^2}$

$= \sqrt{12^2 + 50^2}$

$= \sqrt{16900} = 130$cm = 1.3m (C)

16. $2x + 5y = 11$ ………(i)

$7x + 4y = 2$ ……….(ii)

Using elimination method

Multiply (i) by 7 and (ii) by 2

$14x + 35y = 77$ …………(iii)

$14x + 8y = 4$ ……………..(iv)

Subtract (iv) from (iii)

$27y = 73$

$Y = 73/27$

Substitute the value to y in (i)

$2x + 5(^{73}/_{27}) = 1$

$2x = 11 - {}^{365}/_{27}$

$2x = \dfrac{297 - 365}{27} = \dfrac{-16}{27}$

=> $x = -34/27$

.: $x = -34/27$, $y = 73/27$ (C)

17. $y = 2x^2 + 9x - 35$

By factorizing the L.H.S.

=> $y = 2x^2 + 14x^2 - 5x - 35$

$y = (2x - 5)(x + 7)$

hence, $y < 0 \Rightarrow (2x - 5)(x - 7) \leq 0$.

There are 2 possibilities

(1) $2x - 5 \leq 0$ and $x + 7 \leq 0$ or

(2) $2x - 5 \geq 0$ and $x + 7 \leq 0$

=> (i) $x \leq 5/2$ and $x \geq -7$

=> (ii) $x \geq 5/2$ and $x \leq -7$

But $x > 5/2$ and $x < -7$ is not possible. Therefore, $x \leq 5/2$ and $x \geq -7$

i.e. $-7 \leq x \leq 5/2$ (B)

18. nth term of a G.P. = ar^{n-1}

$a = 9$ ………. …. (i)

$ar^5 = -32/27$ …… (ii)

put (i) in (ii)

$9r^5 = 32/27$

$R^5 = \dfrac{32}{9 \times 27} = \dfrac{2^5}{3^5}$

.: $r = \sqrt[5]{2^5/3^5} = 2/3$

Sum of the first n terms of a G.P =

$\dfrac{a(1 - r^n)}{1 - r}$

.: sum of the first four terms

$= \dfrac{9(1 - [2/3]^4)}{1 - 2/3}$

$= \dfrac{9(1 - 16/81)}{1/3}$

$= 3 \times 9 \times 9\ 65/61 = 21\ 2/3$ (A)

19. $\sin\theta = \sqrt{3}/2 = \theta = 60^0$

.: $\tan(90^0 - \theta) = \tan 30^0 = 1/\sqrt{3}$

$\cos^2\theta = \cos^2 60^0 = (1/2)^2 = ¼$

Hence, $\dfrac{\tan(90^0 - \theta)}{\cos^2\theta}$

$= 1/\sqrt{3} + ¼$

$= 4/\sqrt{3}$

Rationalize to get

$= \frac{4\sqrt{3}}{3}$ (C)

LADOKE AKINTOLA UNIVERSITY

2007 POST UME TEST

MATHEMATICS

1. A regular hexagon is constructed inside a circle of radius 6cm, the area of the hexagon is

   (a) $54cm^2$ (b) $54\sqrt{3}cm^2$ (c) $54\sqrt{2}cm^2$ (d) $54\sqrt{5}cm^2$

2. A solid cylinder of radius 3cm has a total surface area of $36\pi cm^2$

   Find its height.

   (a) 2cm (b) 3cm (c) 4cm (d) 5cm

3. Simplify $\log_{10} 10^3 + \log_{10} 10$

   (a) 61 (b) 3 (c) 5 (d) 4

4. Find the inverse of P under the binary operation * if p*q = p+q-pq where p and q are real number and zero is the identity

   (a) p (b) p-1 (c) $^p/_{p+1}$ (d) $^p/_{p-1}$

5. Find the value of $\int_0^1 \frac{\cos^2\theta - 1}{\sin^2\theta}$

   (a) π (b) $\pi/2$ (c) $-\pi$ (d) $-\pi/2$

6. If dy/dx = 2x – 3 and y = 3 when x = 0, find y in term of x.

   (a) $x^2 - 3x - 3$ (b) $2x^2 - 3x$

   (c) $x^2 - 3x + 3$ (d) $x^2 = -3x$

7. The sum of infinity of the series 1 + 1/3 + 1/9 + 1/27 + 2 is

   (a) 10/3 (b) 11/3 (c) 3/2 (d) 5/2

8. Find the equation of the locus of a point P(x,y) which is equidistant from Q(0,0) and R(2,1)

   (a) 2x + y = 5 (b) 4x + 2y = 5

   (c) 4x – 2y = 5 (d) 2x + 2y = 5

9. Simplify the express $\sqrt{\frac{1 - \cos x}{1+\cos x}}$

   (a) cos x (b) sin x (c) 1 cos x

   (d) $\frac{1 - \cos x}{\sin x}$

SOLUTION

1. **Diagram 4**

   Area of the regular hexagon

   ABCDEF = 6 x area of Δ OAB

Since the triangles are similar

AOB $= 360^0/6 = 60^0$

Area of Δ OAB

$= \frac{1}{2} \times 6 \times 6 \times \sin 60^0$

$= \frac{1}{2} \times 6 \times 6 \times \sqrt{3}/2$

$= 9\sqrt{3}cm^2$

Area of the regular hexagon =

$6 \times 9\sqrt{3}cm^2 = 54\sqrt{3}cm^2$ (B)

2. Total surface area of a cylinder =

$2\pi rh = 2\pi rh = 2\pi r(r + h)$

$\Rightarrow 36\pi = 2\pi \times 3(3+h)$

Divide both sides by 6π

$6 = 3 + h$

$\Rightarrow h = 6 - 3 = 3cm$

The height is 3cm (B)

3. $\log_{10}10^3 + \log_{10}10$

$= 3\log_{10}10 + \log_{10}10 = 3(1) + 1$

$= 4$ (D)

4. Let the inverse of p be p^{-1}.

Then $p*p^{-1} = e$ where 'e' is the identify element

Now $p*p^{-1} = e$

$\Rightarrow p + p^{-1} - pp^{-1} = 0$

(definition of * and identity element being 0)

$P + p^{-1} - (1-p) = 0$

$p^{-1}(1-p) = -p$

$p^{-1} = {}^{p}/_{p-1}$ by dividing both sides

by 1-p (D)

hence the inverse of p = ${}^{p}/_{p-1}$

5. $$\int_0^1 \frac{\cos^2\theta - 1}{\sin^2\theta}\, d\theta$$

from the identify $\sin^2\theta + \cos^2\theta = 1$

we have that

$$\int_0^1 \frac{\cos^2\theta - 1}{\sin^2\theta}\, d\theta$$

$$= \int_0^1 \frac{-\sin^2\theta - 1}{\sin^2\theta}\, d\theta$$

$= \int_0^1 -d\theta = -1/^{n}_{0}$

$= -\pi$ (C)

6. $dy/dx = 2x - 3$

Integrate both sides with respect to x

$y = 2x^2/2 - 3x + c$

$y = x^2 - 3x + c$

At x = 0, y = 3 in *

$3 = 0 - 0 + c$

$\Rightarrow c = 3$

Hence, $y = x^2 - 3x + 3$ (C)

7. $1 + 1/3 + 1/9 + 1/27 + \ldots$

$S_\infty = \dfrac{1}{1 - r}$

From the series, a =1, r = ½

$\Rightarrow S_\infty = \dfrac{1}{1 - 1/3}$

$= S_\infty = \dfrac{1}{2.3} = 3/2$ (C)

8. Equation of the locus of a point p(x,y) which is equidistant from Q (0,0) and R(2,1) is the perpendicular bisector of the line joining Q and R.

Midpoint of QR $= \dfrac{2 + 0}{2} \times \dfrac{1 + 0}{2}$

$= (1, ½)$

Slope of QR $= \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{1 - 0}{2 - 0}$

$= ½$

Therefore, slope of the perpendicular bisector of QR = -2 and a point on it is (1, ½)

Using the formula $\dfrac{y_2 - y_1}{x_2 - x_1} = m$

The equation of the perpendicular bisector is $\dfrac{y - ½}{x - 1} = -1$

$\Rightarrow y - ½ = 2x + 2$

Multiply through by 2

$2y - 1 = 4x - 4$

$4x + 2y = 5$ (B)

9. $\sqrt{\dfrac{1 - \cos x}{1 + \cos x}}$

$= \sqrt{\dfrac{1 - \cos x}{1 + \cos x} \times \dfrac{1 - \cos x}{1 - \cos x}}$

$= \sqrt{\dfrac{(1 - \cos x)^2}{1 + \cos x}}$

$= \sqrt{\dfrac{(1 - \cos x)^2}{\sin^2 x}}$

$= \dfrac{1 - \cos x}{\sin x}$ (D)

LADOKE AKINTOLA UNIVERSITY
2008 POST UME TEST
MATHEMATICS

1. If x + 1 is a factor of $2x^3 + 3x^2 + kx+4$, find the value of k..
(a) - 5 (b) 5 (c) -4 (d) 4 (c) 3

2. Obtain the maximum value of the function $f(x) = x^3 - 12x + 4$
(a) 2 (b) 5 (c) -2 (d) 2 (e) 27

Use the table below to answer question 13

No of children	0	1	2	3	4	5	6
No of families	7	11	6	7	7	5	3

3. Find the mean of the distribution
(a) 2 (b) 3 (c) 2.5 (d) 3.5 (e) 4

4. Let A = (1,3,7,8) and B = (2,4,7,9). The element which shows that the set A and B are not disjoint is ……
(a) 3 (b) 7 (c) 4 (d) 2 and 1 (e) 8

5. Let $\binom{n}{r} = \dfrac{n!}{r!(n-r)}$

Evaluate $\binom{7}{3} + \binom{5}{2}$
(a) 35 (b) 75 (c) 45 (d) 55 (e) 30

6. Find the equation of he straight line through (1,2) and 3,8)
(a) y=x-3 (b) 2y=3x+1
(c) y=3x-1 (d) 2y = 3x – 1

7. Let $f(x) = \dfrac{3x + 1}{X + 2}$. Find $\int f(x)dx$
(a) 5x - 3ln/x + 2/+K

(b) 3x – 5In/x+2+K

(c) 5x/3In/x + 2/+K

(d) 3x/5In/x + 2/+K

(e) E.K

8. If $\log_2(3y^2 + 8y - 1) = 1$, evaluate Y.

(a) y = 1/3 or -3 (b) y=-1/3 or 3

(c)y=-1/3 or -3 (d) y = 1/3 or 3

(e) y = 3 (twice)

SOLUTION

1. Let $p(x) = $ of $2x^3 + 3x^2 + kx+4$. Since $x + 1$ is a factor, by factor theorem, $p(-1) = 0$

$\therefore p(-1) = 2(-1)^3 + 3(-1)^3 + k(-1) + 4 = 0$

$\Rightarrow -2 + 3 - k + 4 = 0$

$\Rightarrow 5 - k = 0$

$\Rightarrow k = 5$

2. $f(x) = x^3 - 12x + 4$. At turning point, $d/dx \ f(x) = 0$

$\Rightarrow 3x^2 - 12 = 0$

$X^2 = 4$

$\Rightarrow x = \pm\sqrt{4} = \pm 2$

At $x = 2$, $f(2) = 2^3 - 12(2) + 4$

$= 8 - 24 + 4 = -12$

At $x = -2$, $f(-2) = (-2)^3 - 12(-2) + 4$

$= -8 + 24 + 4 = 20$

Hence, the maximum value of the function is 20

3. Mean $= \dfrac{\sum fx}{\sum f}$

$= \dfrac{11 + 12 + 21 + 28 + 25 + 18}{46}$

$= 2.5$ (C)

4. Reason: $A \cap B = (7)$ (B)

5. $\binom{7}{3} + \binom{5}{2}$

$= 71/3{:}4 + 51/2{:}3$

$= \dfrac{7 \times 6 \times 5}{3{:}4} + \dfrac{5 \times 4}{2}$

$= 35 + 10 = 45$ (C)

6. Using $\dfrac{y = y_1}{X = x_1} = \dfrac{y_2 - y_1}{x_2 - x_1}$

$\Rightarrow \dfrac{y-2}{X-1} = \dfrac{8-2}{3-1}$

$\dfrac{y-2}{X-1} = \dfrac{6}{2}$

$\Rightarrow y - 2 = 3(x - 1)$

$y - 2 = 3x - 3$
$\Rightarrow y = 3x - 1$ (C)

7. $\int \frac{3x + 1}{x + 2} dx$
$= \int (3 - \frac{5}{x+2})dx$
$= \int 3dx - \int \frac{5}{x+2} dx$
$= 3x - 5\ln(x + 2) + k$ (B)

8. $\log_2(3y^2 + 8y - 1) = 1$
$\Rightarrow 3y^2 + 8y - 1 = 2^1$
$\Rightarrow 3y^2 + 8y - 3 = 0$
$3y^2 + 9y - y - 3 = 0$
$3y(y+3) - 1(y+3) = 0$
$(3y - 1)(y + 3) = 0$
$3y - 1 = 0$ or $y + 3 = 0$
$\Rightarrow y = 1/3$ or -3 (A)

LADOKE AKINTOLA UNIVERSITY
2009 POST UME TEST
MATHEMATICS

1. Express p in terms of q if $\log_4 p + 2\log_4 q = 4(16/q)^2$
(a) $(16/q)^2$ (b) $(4/q)^2$ (c) $(q/16)^2$
(d) $(q/4)^2$
2. Find the possible values of p if the expression $4x^2 - 3px + 3p$ leaves a reminder 10 when divided by x -p
(a) 2 (b) 3 (c) 4 (d) 5
3. If p and q are the roots of the equation $2x^2 - 5x - 7 = 0$, find the value of $\beta/a + a/\beta$
(a) 3/11 (b) -3/14 (c) 2/14 (d) -2/14
4. Find the locus of a point that is equidistance from the points (1,2) and (3,8)
(a) $y = \frac{1}{3(17-x)}$ (b) $y = \frac{1}{3(x+13)}$
(c) $y = \frac{1}{3(x+8)}$ (d) $y = \frac{1}{3(11-x)}$
5. A bag contains 5 black balls and 3 red balls. Two ball are picked at random without replacement. What is the probability that a black ball and a red ball are picked?
(a) 15/28 (b) 13/28 (c) 5/14
(d) 11/12
6. A rectangular hexagon is constructed inside a circle of diameter 12cm. calculate the area of the hexagon.
(a) $36\pi cm^2$ (b) $36\sqrt{3}cm^2$
(c) $54\sqrt{3}cm^2$ (d) $54\pi cm^2$
7. If the three consecutive terms of a GP are n – 2, n and n + 3, find the common ratio.
(a) ¼ (b) ½ (c) ¾ (d)1 ½

8. In a triangle ABC, if the AB = 15cm and angle ABC = 45^0 while angle ACB = 60^0, find the length of line AC

(a) 15√6cm (b) 10√6cm

(c) 20√6cm (d) 5√6cm

9. Find the derivative of the function

$y = x^2 (4x + 3)$ at the point x = 2

(a) 60 (b) 50 (c) 40 (d) 30

SOLUTION

1. $Log_4\ p + 2\ log_4\ q = 4$

$Log_4\ p + log_4\ q^2 = 4$

$Log_4 pq^2 = 4$

$=> pq^2 = 4^4$

$= = 4^4/q^2 = (4^2/q) = (16/q)^2$ (A)

2. Let $f(x) = 4x^2 – 3px + 3p$

Then, when $f(x)$ is divided by x – p, the remainder is $f(p)$

.: $f(p) = 4p^2 – 3p \times p + 3p = 10$

$4p^3 – 3p^2 + 3p = 10$

$P^2 + 3p – 10 = 0$

$P^2 + 5p – 2p – 10 = 0$

P(p+5) – 2(p+5) = 0

(p+5) (p-2) = 0

=> p + 5 = 0 or p -2 = 0

=> p = 15 or 2 (A)

3. $2x^2 – 5x + 7 = 0$

The general quadratic equation is of the form $ax^2 + bx + c = 0$

From this, we have

$X^2 + b/a\ X + c/a = 0$

If α,β are roots of the quadratic, then $\alpha + \beta = -b/a$

$\alpha\beta = c/a$

.: For $2x^2 – 5x + 7 = 0$

a = 2; b = -5; c = +7

.: $\alpha + \beta = 5/2$, $\alpha\beta = + 7/2$

Hence, $\alpha/\beta + \beta/\alpha = \dfrac{\alpha^2 + \beta^2}{\alpha\beta}$

$= \dfrac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$

$\dfrac{(5/2)^2 – 2(+7/2)}{-7/2} = \dfrac{25/4 – 7}{-7/2}$

$= \underline{25 – 28} \times \underline{-2} = \underline{3} \times \underline{-2}$

4 7 4 7

= 3/14 (A)

4. Let the point be p(x,y)

Distance between p(x,y) and (1,2)

$= \sqrt{(x-1)^2 + (y-2)^2}$

Distance between p(x,y) and (3,8)

$= \sqrt{(x-3)^2 + (y-8)^2}$

$.: \sqrt{(x-1)^2 + (y-2)^2} = \sqrt{(x-3)^2 + (y-8)^2}$

$=> (x-1)^2 + (y-2)^2 = (x-3)^2 + (y-8)^2$

$X^2 - 2x + 1 + y^2 - 4y + 4 = x^2 - 6x + 9 + y^2 - 16y + 6x - 2x + 16y - 4y + 1 + 4 = 9 + 64$

4x + 12y = 73 – 5

4x + 12y = 68

Divide through by 4

X + 3y = 17

3y = 17 – x

Y = 1/3 (17-x) (A)

5. Let pr (B) be the probability of picking a black ball and pr (R) be the probability of picking a red ball

Pr (B & R) = Pr (B/R) or pr (R/B)

= (5/8 x 3/7) + (3/8 x 5/7)

= 15/56 + 15/56

= 30/56 = 15/28 (A)

6. **diagram 5**

Area of the hexagon = 6 x area of ΔOAB. Using the diagram above

$\theta = 360^0/6 = 60^0$

Area of ΔOAB = ½ x 6 x 6 sin 60^0

= 18 x √3/2 = 9√3cm^2

.: Area of the hexagon = 6 x 9√3cm^2

= 54√3cm^2 (C)

7. Since n -2, n and n + 3 are consecutive, the common ratio r;

$r = \frac{n}{n-2} = \frac{n+3}{n}$

solving $\frac{n}{n-2} = \frac{n+3}{n}$

$n^2 = (n + 3)(n - 2)$

$n^2 = n^2 - 2n + 3n - 6$

$n^2 = n^2 + n - 6$

0 = n – 6

=> n = 6

Hence, the common ratio is 6/4 = 3/2 (D)

8. **Diagram 6**

Using sine rule b/sinB = c/sinC

$b/\sin 45^0 = 15/\sin 60^0$

$b = \frac{15 \times \sin 45^0}{\sin 60^0} = (15 \times \frac{1}{\sqrt{2}}) + \frac{\sqrt{3}}{2}$

$= 15 \times \frac{1}{\sqrt{2}} \times \frac{2}{\sqrt{3}}$

$= 30/\sqrt{6} \times \sqrt{6}/\sqrt{6}$

$= \frac{30\sqrt{6}}{6} = 5\sqrt{6}$

9. $y = x^2 (4x + 3)$

$= 4x^3 + 3x^2$

$dy/dx = 12x^2 + 6x$

Then, at x = 2

$dy/dx = 12(2)^2 + 6(2)$

$= 12(4) + 6 (2)$

$= 48 + 12$

$= 60$ (A)

LADOKE AKINTOLA UNIVERSITY

2010 POST UME TEST

CHEMISTRY

I. A metallic element X forms an on X2 what will be the formula of tetraoxosulphate of x

a) $X(S0_4)_3$ b)X SO_4 c) $X_2(SO_4)$ d) X_3S0_4

2. When air is passed through alkaline pyragallol. the component of air absorbed by the pyragallol is

a) Oxygen b) neon c) nitrogen d) argon

3. When water drops are added to calcium carbide in a container and the gas produced is passed through jet and lighted, the resultant flame is called an

a) Oxyethane flame b) Oxyethyleno flame c) Hydroxide flame d) Oxyacetylene flame

4. If the relative formula mass of a hydroxide $M(OH)_3$ is $78gmo1^{-1}$ what is the relative atomic mass, M?

a) 61 b) 59 c) 30 d) 27

5. An ion with a single negative charge becomes an atom by

a) Losing an electron b) losing a neutron c) gaining a neutron d) gaining an electron

6. $CH_3COOH + CH_3OH$ — $CH_3COOCH_3 + H_20$.

The reaction represented by the equation above is

a) neutralization reaction b) esterification reaction c) de esterification reaction d) hydrolysis reaction

7. The metal extracted from Bauxite is

a) Calcium b) Magnesium c) Aluminium d) Copper

8. At atom of an element represent by X with A 17 and Z = 8. How many neutron does X contains?
   a) 3 neutrons b) 5 neutrons c) 4 neutrons d) 9 neutrons
9. What type of oxide is aluminium oxide?
   a) Basic oxide b) acidic oxide c) neutral oxide d) amphoteric oxide
10. Excess ethanol react with acidified $KMnO_4$ to form
    a) Ethene b) Ethanoic acid c) Ethylethanoate d) Ethane
11. What is the value of X in the reaction below'?
    a) 234 b) 238 c) 230 d) 239
12. Two gas cylinder cc $^{238}_{92}U$ → $^{X}_{90}Th$:etylene respectively one test, which can be used to distinguish between them is by
    a) Passing each gas through dilute potassium permanganate sodium b) Pass each gas through ammonical copper 1 chloride solution c) Passing each through bromide water d) Treating each gas catalytically with excess hydrogen gas
13. Brass and bronze are both metallic alloys, which of the following constituents is common to be alloys
    a) Tin b) Zinc c) Copper d) Lead
14. An organic compound has the empirical formula CH,O. What is its molecular formula if its vapour density is 90? (H = 1, C =12.0 = 16)
    a) C4H304 b) C12H30 c) C4H1005 d) C6H1O6
15. A technique that can be used to show that chlorophyll pigment in a mixture of chemical compound and not a single coloured
    a) Hydrolysis b) crystallization
    c)chromatography d) sublimation
16. Which of the following statements is not true as we move from left to right along the periodic table?
    a) Atomic mass of elements increase
    b) Electro positive character of elements increase
    c) Atomic number of element increases
    d) Number of electrons in the outermost orbits of elements increases
17. What mass of a divalent metal. M (atomic mass 40) would react with excess hydrochloric acid to liberate $224cm^3$ of dry hydrogen gas measured as s.t.p.?
    a) 8.0gb) 0.04g c) 0.4g d) 1.0g (GMV = 22.4dm3)
18. Which of the following is an acid salt?
    a) $NaHSO_2$ b) $Na2SO_4$
    c) CH_3COONa d) Na_2S
19. An element with atomic number twelve is likely to be
    a) Electrovalent with a valency of the b) Electrovalent with a valency of 2 c) Covalent with a valency of 2 d) Covalent with a valency of 4
20. The approximate volume of air containing 10cm3 of oxygen is
    a) 10cm b) 20cm c) 50cm d) 100cm

21. Duralumin consist of aluminium, copper and a) Zinc and gold b) lead and manganese c) nickel and silver d) manganese and magnesium
22. Sodium hydroxide solution can be conveniently stored in a container made of a) lead b) zinc c) aluminium d) copper
23. The energy change accompanying the addition of an electron to a gaseous atom is called
    a) First ionization energy

b) Second ionization energy

c) Electron affinity

d) Electron activity.

24. 50cm3 of hydrogen is sparked with 20cm3 of oxygen at $100^{0}C$ and atmosphere. The total volume of the residual gases is

a) $50cm^3$ b) $10cm^3$ c) $40cm^3$ d) $30cm^3$

25. The molarity of 2% by weight of aqueous solution of H_2S0_4 (molecular weight of 98)

a) 3.55 b) 2.55 c) 0.02 d) 0.55

26. Alkanoic acids have volatility compared with alkanols because

a) They are more polar than alkanol b) They have two oxygen atoms while alkatiols have one

c) They form two hydrogen bonds while alkanols form one

d) They form hydrogen bonds while alkanols do not.

27. How many faradays of electricity are required to deposit 0.20 mole of nickel if 0.10 faraday electricity deposited 2.9.8g of nickel during electrolysis its aqueous solution? (N= 58 if 96500 mol^{-1})

a) 0.20 b) 0.39 c) 0.40 d) 0.50

28. In the process of silver plating a metal M the metal M is the

a) Anode and a direct current is used b) Cathode and an alternating current is used

c) Anode and an alternating current is used d) Cathode and direct current is used.

29. When a gas is compressed at very low temperature

a) Its density decreases b) It liquefies c) Its temperature decreases d) Temperature increases

30. Oxidation of concentration hydrochloric acid with manganese (IV) oxide liberates a gas used in the

a) Manufacture of toothpaste

b) Treatment of goiter

c) Vulcanization of rubber

d) Sterilization water

Answer key

1: B 7. C 13.C 19.B 25C

2. A 8. D 14.D 20.C 26.C

3. D 9. D 15.C 21.D 27.C

4. A I0.B 16.B 22.D 28.D

5. A IIA.B 17.C 23.C. 29.B

6. B I2.B 18.A 24.A 30.D

EXPLANATION TO ANSWERS

I The metal must be divalent, i.e. having 2- electrons in its outermost shell to donate to the sulphate, $SO^2{}_4$ to give XSO_4 (B)

2 Alkaline pyragallol is used to absorb oxygen (A)

3 CaC_2 + 2H,O → C2H, + $Ca(OH)_2$

Ethyne (ethylene) produced when passed through oxygen give oxyacetylene flame, used for welding (D)

4. $M(OH)_3$ = 78gmo1^{-1}

$M+(16+1)_3 = 78$

M+(17x3) = 78

M=78—51 = 27(D)

5. E.g. X → X + e (loss of electron) (A)

$CH_3COOH + CH_3OH \xrightarrow{H^+} CH_3COOCH_3 + H_2O$

6.

The reaction is acid catalysed and reversible produce methylethanoate (ester) in the process called esterification (B)

7. Bauxite ($A1_2O_32H_2O$) is an ore of aluminium (C)

8. $^{A}_{Z}X$ when A = 17 mass number

X = 8 atomic number

Atomic No. = no. of proton + no. of neutron = 17--8 = 9 (D)

9. Aluminium form oxide called amphoteric. They exhibit both acidic oxide and basic oxide characteristics (D).

10. This is oxidation reaction, ethanol (a primary alcohol) gives ethanoic acid.

$C_2H_5OH \xrightarrow{[O]}$ CH_3COOH (ethanoic acid) (B)

11. $^{238}_{92}U \longrightarrow {}^{x}_{90}Th + {}^{4}_{2}He \ (X \ is \ 234) \ (A)$

12. Ethylene ($C2H_4$) and acetylene (C_2H_2). All alkynes with a terminal triple bond undergo substitution reaction with CuCl to give reddish brown colouration of Cu_2C_2 (B).

13. Brass & Bronze are alloys they both contain copper (C).

14. Vapour density = ½ x relative molar mass Relative molar mass 2 x 90

= 180

$(CH_2O)_n$ = 180

$[(12 \times 1)+ (2) + 16)]_n$. 180

30_n = 180

n =6

$(CH_2O)_6 = C_6H_{12}O_6$ (D)

15. Chromatography is a separation technique used to separate component, mostly coloured, depending on their rate of migration (C)

16. Although electropositivity decreases across the period, while electronegativity increase (B)

17. M + 2HCl → MC_{l2} + H_2

1 mole: 2 mole 1mole : 1mole

No. of mole = $\frac{0,224.m^4}{22.4dm}$

= 0.0lmole of hydrogen gas

Since 1 mole of H2 is liberated by 1mole of M 0.0lmole of H_2 will liberate 1mole of M Mass of M = 40x0.01 = 0.4g (C)

18. Acid salt are produced as a result of incomplete displacement of hydrogen, e. g. $NaHSO_4$, $NaHCO_3$ (A)

19. An element with atomic number 12 is metallic divalent and a reducing agent, ionize by losing 2 electrons to combine with non-metals to form electrovalent bond (B)

20. Oxygen is 21% of oxygen

% oxygen = $\frac{\text{Vol of oxygen}}{\text{Total Vol of air}} \times 100$

21%= $\frac{10cm^3}{\text{Total volume of air}} \times 100$

Total vol. of air = $\frac{10cm^3}{21\%} \times 100\%$

21. Duralumin is an alloy of aluminium. copper, magnesium and manganese (D).

22. Zinc and aluminium readily dissolve in sodium hydroxide to form sodium zincate(1I) and sodium aluminate(III), respectively. Sodium hydroxide do not react with lead. Copper, being fairly far down the activity series, has no action with alkalis (D)

23. The first ionization energy is the energy required to remove one electron from each atom is a mole of gaseous atoms, producing one mole of gaseous ion with a position charge while electron affinity is the ability of an atom in gaseous form to attract electron toward itself (C)

$2H_2 + O_2 \rightarrow. 2H_2O$

Reaction mole 2moles 1mole 2 moles

Reacting atom 50cm2 20cm3 -

Reaction ratio 40cm 20cm 40cm

Residual gas 10cm3 - 40cm3

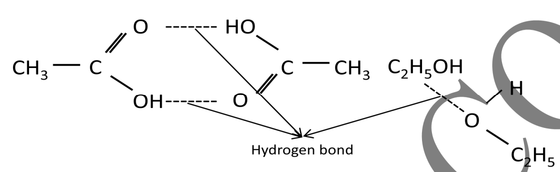
Residual gas = unreacted hydrogen + formed water

= 10 + 40 = 50cm (A)

25. 2g of H2S04 is in 1000cm3 of water Mass conc. =2g/dm3

molar conc. = mass conc./ molar mass 2/98 0.O2gldm^{-3}

26. Alkanoic acid is less volatile because it is more stable compact to alkanol, it form dimmers that is two molecule are bonded at 2-site by hydrogen bond



27 Ni $\longrightarrow$ Ni^{2+} + $2e^-$

At step 1 mole 2F

No. of mole $= \frac{29.8}{58} = 0.513$mole

0.Ol53mole of Nickel will be deposited by 0. IF

0.2mole of Nickel will be deposited by $\frac{0.1 * 0.2}{0.513}$

= 0.389F

0.4F (C)

28. As the current is passed through the cell, the plating metal dissolve at the anode, and the ion produced migrate to the cathode, where they are discharge and deposited as a layer on the object by direct current (D).

29. When air is compressed, it becomes liquefied (B)

30 $MnO_2 + 4HCl \xrightarrow{Heat} MnCl_2 + 2H_2O + Cl_2$

Chlorine is used in the treatment of water or for sterilization. (D)

LAKINTOLA UNIVERSITY OF TECHNOLOGY 2007 POST LIME TEST

I. The relative strengths of weak acids can be seen by a comparison of their

a) pH value b) dissociation constants

c) hydrogen ion concentrations

d) molarities

2. Which of the following gases has the same volume under the same condition as 16g of sulphur (IV) oxide? (H = 4, C = 12, N= 14, O = 16, S = 32, Cl = 35)

a) 8g of hydrogen

b) 5.5g of carbon(VI) oxide c) 11.0g of carbon (IV) oxide d) 5g of nitrogen

OH

3. CH_3—CH—CH_2—CH_2—NH_2. The two functional groups in the above compound are

a) Amino and carbonyl b) Carbonyl and hydroxyl c) Hydroxyl and carboxyl

d) Hydroxyl and amine

4. F^{-1} would have a larger radius than fluorine atom, because of

a) Addition of an extra electron in F^{-1}

b) The ion was obtained after fluorine lost valence electron

c) The effective nuclear charge of F^{-1} is greater than that of F^{-1}

d) F is an ion while F is not.

5. 0.1 Faraday of electricity was passed through a solution copper(II) sulphate. The maximum weight of copper deposited on the cathode would be

a) 63.Og b) 32.Og c) 6.lg d) 3.2g

6. What is the concentration of H^+ ion in mole's per dm3 of the solution of pH 4.398?

a) 4.0×10^{-2} b). 4.0×10^{-3} c) 4.0×10^{-4} d) 4.0×10^{-6}

7. When 50cm3 of a saturated solution of a sugar (molar mass 342.0g) at 40°C was evaporated to dryness 31.2g of dry solid was obtained. The solubility of sugar at 40°C is

a) 10.0 moles cm^3 b) 7.Omoles dm^{-3} c) 3.5moles dm^3 d) 2.Omoles dm^{-4}

8. When H2S is passed into a solution of iron (III) chloride, the colour changes form yellow to green. This is because

a) H_2S is reduced to S

b) H_2S ion. are oxidized by Fe^{2+}

c) Fe_2^+ ions are oxidized by Fe^{2+} ions

d) Fe^{2+} ions are reduced to Fe^2 ions

9. The oxide that remains unchanged when heated in hydrogen is

a) CuO b) Fe_2O_3 c) PbO_2 d) ZnO

ANSWER KEY

1 C2 C 3D 4 A 5 D 6- 7D

8D 9A

EXPLANATION TO ANSWER

1. The characteristic properties of an acid in solution are due to the presence of hydrogen ion. Weak acids are only partially ionized in water. Such acid solution have a low concentration of hydrogen. Dissociation, constant K, is used to determine the strength of the end (B)

2 16g of SO_2 $\longrightarrow$ no. of mole = $\frac{16}{64}$ = 0.25

8g of H_2 $\longrightarrow$ no. of mole = $\frac{8}{2}$ = 4

5.5g of CO_2 $\longrightarrow$ no. of mole = $\frac{5.5}{44}$ = 0.125

11.0g of CO_2 $\longrightarrow$ no. of mole = $\frac{11.0}{44}$ = 0.25

5g of N_2 $\longrightarrow$ no. of mole = $\frac{5}{28}$ = 0.14 (C)

3 CH_3 –CH(OH) – CH_2 – CH_2NH_2 (Hydroxyl & amine) (D)

4. The ionic radii of negative ions are greater than the corresponding atomic radii, this is because a negative ion is formed by adding electron to the outermost shell, thus making it bigger (A)

5. $Cu^{2+} + 2e^{-}$ — Cu

1mole 2F

2F would deposit lmole of Cu

iF would deposit ½ mole of Cu

0.IF would deposit (½ x 01) mole of Cu

= 0.05mole of copper

mass = 0.05 x 63.5

= 3.2g(D)

6. pH -log $[H^+]$

4.398 = -log $[H^+]$

$[H^+]$ = antilog (-4.398) [do it yourself]

7. Sugar + solution = sugar solution

$50cm^3$

At 40°C solubility = 34.2g in 50g water Solution contain 34.2g

342g

No. of mole = 34.2g/mol

= 0.lmole

Solubility in mol/dm = $\frac{10.1}{50/1000}$

= 0.1 x 20

= $2mol/dm^3$ (D)

8. $H_2S + 2FeCl_3 \rightarrow FeCl_2 + 2HCl + S$

-2 +3 +2 0

Hydrogen sulphide reduces a brownish yellow solution of iron(III) chloride to a green solution of iron(II) chloride. The hydrogen sulphide itself is oxidized to sulphur and hydrogen chloride. Iron(III) to iron(II) is gain of electron, the process is reduction i.e. Fe^{3+} is reduced to Fe^{2+} (D)

9. Copper is below hydrogen in activity series. Therefore, copper can not displace hydrogen (A)

LADOKE AKINTOLA UNIVERSITY OF TECHNOLOGY 2009 POST UME TEST

1. The formula of the compound formed in a reaction between a trivalent metal, M and a tetravalent non-metal X is

a) MX b) M_3X_4 c) M_4X_3 d) M_3X_2

2. One mole of propane is mixed with five moles of oxygen. The mixture is ignited and the propane burns completely. What is the volume of the products at s.t.p.?

a) $112.0dm^3$ b) $67.2dm^3$ c) $56.0dm^3$ d) $4.8dm^3$

[G. M. V. = 22.4 dm3mol']

3. How many unpaired electrons are there in the electronic configuration of an element with atomic number 8?

a) 2 b)3 c)4 d)5.

4. The secondary valence is called

a) Oxidation state b) Lewis base c) coordination number d) chelates

5. Which of these is not a hydroscopic salt?

a) Calcium oxide b) magnesium chloride

c) copper (II) oxide d) sodium trioxonitrate(V)

6. The order of decreasing s-character of Sp,Sp^3& Sp^2 is

a) Sp, Sp^2, Sp^3 b) Sp^3, Sp^2, Sp c) Sp^2,Sp, Sp^3 d) Sp, Sp^3, Sp^2

7. Brass and Bronze are both metallic alloys. Which of the following constituents is common to both alloys?

a) Lead b) Copper c) Tin d) Zinc

8. The major process of manufacturing caustic soda on industrial scale is

a) Reduction of brine

b) Fusion of sodium and hydroxide

c) Electrolysis of brine

d) Concentration of brine

9. Which of these compounds is not used as a fertilizer?

a) Sodium hydroxide

b) carbamide c) ammonium trioxonitrate

d) potassium tetraoxosulphate VI

10. How many molecules are there in 5.6dm3 of ammonia at s.t.p.? G.V.M. = 22.4dm3 at s.t.p. NA = 6.02 x $l0^{23}$moV'

a) 1.50 x 10^{20} b) 1.50 x 10^{21} c) 1.50 x 10^{22} d) 1.50 x 10^{23}

Answer key

1.C	4.C	7.B	10.D
2B	5.B	8.C	
3.A	6.A	9.A	

EXPLANATION TO THE QUESTION

1. Metal M^{3+} (Trivalent)

X^{-4} (Tetravalent)

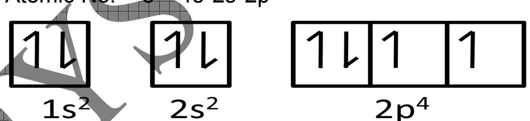
M is having 3-electrons in its outermost shell. It will donate 3-electrons in 4 folds to form an octet with X in 3-folds, to form M_4X_3 (C)

2. $C_3H5 + 5O_2 \rightarrow 3C02 + 4H_2O$ 1 mole 5mole 3mole 4mole

Vol of CO_2 = No. of mole x 22.4 at stp

3 x 22.4 = 67.2dm_3 (B)

3 Atomic No. = 8 = $1s^2 2s^2 2p^4$



The electronic configuration gives 2-unpaired electron (A)

4. The secondary valency is called co-ordination number (C)

5. $MgCl_2$ is not hydroscopic but deliquescence (B)

6	Sp – 50%	S – character & 50%	p-character
	Sp^2- 33%	S – character & 66.6%	p-character
	Sp^3 – 25%	S – character & 75%	p-character

Order of decreasing sp > sp^2> sp^3 (A)

7. Brass & Bronze are both metallic alloy having copper as their common constituents (B)

9. Sodium hydroxide is an alkali, it is not used in fertilizer manufacturing. Carbamide give the soil nitrogen. ammonium trioxonitrate(V) give the soil ammonium and K_2SO_4 give sulphur (A)

10. At stp 22.4dm^3 of ammonia have 6.02 x 10^{23} molecule

5.6dm^3 of ammonia will have

$\frac{6.02*10^{23}\ molecule\ 5.6}{22.4}$ = 1.505 x 10^{23} molecule (D)

